Swimming Filaments!!

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1 Micropolar media

In this section we consider micropolar media whose microstates take values in the group of rigid transformations, which we choose to represent as

$$SE(3) = \left\{ \begin{bmatrix} Q & r \\ 0 & 1 \end{bmatrix} \in GL(4) \middle| Q \in SO(3), r \in \mathbb{R}^3 \right\}. \tag{1}$$

The state of such a medium can be decomposed Components of tensors on $SE(3)$ will have Greek indices,

$$T^{\alpha}_{\beta\gamma}, \chi^{\mu}_{\nu}, \text{ etc} \ldots, \tag{2}$$

which take values in $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$,

$$T^{\alpha}_{\beta\gamma}\chi^{\beta}_{\alpha} := \sum_{\beta,\alpha=0}^{5} T^{\alpha}_{\beta\gamma}\chi^{\beta}_{\alpha}. \tag{3}$$

Spacetime coordinates $s = (s^0 = t, s^1, \ldots, s^p)$, consisting of a single timelike coordinate $s^0 = t$ and p spacelike material coordinates $(s^1, \ldots, s^p)$ are used throughout. Components of tensors on spacetime will have latin indices,

$$F^{a}_{bc}e^{m}_{n}, \text{ etc} \ldots, \tag{4}$$

which take values in $\mathbb{Z}_{p+1} = \{0, \ldots, p\}$

$$F^{a}_{bc}e^{b}_{a} := \sum_{a,b=0}^{p} F^{a}_{bc}e^{b}_{a}. \tag{5}$$

The state of a micropolar medium on the spacetime manifold $S \subset \mathbb{R}^{p+1}$ is a map

$$\rho : S \to SE(3) \tag{6}$$

which assigns a microstate to each point in S. Kinematical constraints due to the geometry of $SE(3)$ are discussed in 1.1. We introduce a sigma-model action in 1.2 and outline conditions for extremality. Explicit equations of motion for the Kirchoff rod, an example of a $p = 1$ micropolar material, are found in 1.3.

1.1 Kinematics and geometry

1.1.1 Spacetime

Let $S \subset \mathbb{R}^{p+1}$ be a compact and simply connected domain with coordinates

$$(s^0, s^1, \ldots, s^p) = (t, s^1, \ldots, s^p) \tag{7}$$

and Minkowski metric with signature $(+, -, \ldots, -)$

$$\left\langle \mathrm{d}s^i \middle| \mathrm{d}s^j \right\rangle = \begin{cases} -1 & \text{if } i = j > 0 \\ 1 & \text{if } i = j = 0 \\ 0 & \text{if } i \neq j \end{cases} \tag{8}$$

Choose your favorite matrix representations of the Lie algebras $\mathbb{R}^3$, $\mathfrak{so}(3)$, and $\mathfrak{se}(3)$:

1.1.2 $\mathbb{R}^3$

- Basis:

$$e_0, e_1, e_2 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix} \tag{9}$$

- Inner product:

$$\langle a|b\rangle_{\mathbb{R}^3} := \mathrm{tr}\big(a^T b\big) = a^T b \tag{10}$$

- Metric components:

$$\delta_{\alpha\beta} := \langle e_\alpha|e_\beta\rangle_{\mathbb{R}^3} \tag{11}$$

- Lie bracket:

$$[a, b]_{\mathbb{R}^3} := a \times b \text{ (cross product)} \tag{12}$$

- Structure constants:

$$\varepsilon_{\alpha\beta\gamma} := \langle e_\alpha|[e_\beta, e_\gamma]\rangle \tag{13}$$

1.1.3 $\mathfrak{so}(3)$

- Basis

$$\hat{e}_0, \hat{e}_1, \hat{e}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{14}$$

- Inner product

$$\left\langle \hat{a} \middle| \hat{b} \right\rangle_{\mathfrak{so}(3)} := \mathrm{tr}\left(\hat{a}^T \hat{b}\right) \tag{15}$$

- Metric components

$$\langle \hat{e}_\alpha | \hat{e}_\beta \rangle = 2\delta_{\alpha\beta} \tag{16}$$

- Lie bracket

$$\left[\hat{a}, \hat{b}\right]_{\mathfrak{so}(3)} := \hat{a}\hat{b} - \hat{b}\hat{a} \tag{17}$$

- Structure constants

$$\langle \hat{e}_\alpha | [\hat{e}_\beta, \hat{e}_\gamma] \rangle = 2\varepsilon_{\alpha\beta\gamma} \tag{18}$$

- Action on $\mathbb{R}^3$

$$\langle e_\alpha | \, \hat{e}_\beta \, | e_\gamma \rangle = \varepsilon_{\alpha\beta\gamma} \tag{19}$$

1.1.4 $\mathfrak{se}(3)$

- Basis

$$\phi_0, \phi_2, \phi_4 = \begin{bmatrix} \hat{e}_0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \hat{e}_1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \hat{e}_2 & 0 \\ 0 & 0 \end{bmatrix} \tag{20}$$

$$\phi_1, \phi_3, \phi_5 = \begin{bmatrix} 0 & e_0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_2 \\ 0 & 0 \end{bmatrix} \tag{21}$$

- Inner product

$$\langle a|b\rangle_{\mathfrak{se}(3)} := \mathrm{tr}\big(a^T b\big) \tag{22}$$

- Covariant metric components

$$g_{\alpha\beta} := \langle \phi_\alpha | \phi_\beta \rangle \tag{23}$$

$$g_{\alpha\beta} = \frac{3 + (-1)^\alpha}{2}\delta_{\alpha\beta} = \begin{cases} 2\delta_{\alpha\beta} & \text{if } \alpha, \beta \in \{0, 2, 4\} \\ \delta_{\alpha\beta} & \text{if } \alpha, \beta \in \{1, 3, 5\} \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

- Contravariant metric components

$$\big[g^{\alpha\beta}\big] := [g_{\alpha\beta}]^{-1} \text{ (matrix inverse)} \tag{25}$$

$$g_{\alpha\beta} = \frac{2}{3 + (-1)^\alpha}\delta_{\alpha\beta} = \begin{cases} \frac{1}{2}\delta_{\alpha\beta} & \text{if } \alpha, \beta \in \{0, 2, 4\} \\ \delta_{\alpha\beta} & \text{if } \alpha, \beta \in \{1, 3, 5\} \\ 0 & \text{otherwise} \end{cases} \tag{26}$$

- Lie bracket

$$[\mu, \nu]_{\mathfrak{se}(3)} := \mu\nu - \nu\mu \tag{27}$$

- Structure constants

$$\begin{aligned} c_{\alpha\beta\gamma} &:= \langle \phi_\alpha | [\phi_\beta, \phi_\gamma] \rangle \\ c_{2\alpha,2\beta,2\gamma} &= 2\varepsilon_{\alpha\beta\gamma} \\ c_{2\alpha+1,2\beta,2\gamma+1} &= c_{2\alpha+1,2\beta+1,2\gamma} = \varepsilon_{\alpha\beta\gamma} \\ c_{2\alpha,2\beta,2\gamma+1} &= c_{2\alpha,2\beta+1,2\gamma} = c_{2\alpha,2\beta+1,2\gamma+1} = c_{2\alpha+1,2\beta,2\gamma} = c_{2\alpha+1,2\beta+1,2\gamma+1} = 0 \end{aligned} \tag{28}$$

Where $\alpha, \beta, \gamma \in \{0, 1, 2\}$ and $\varepsilon_{\alpha\beta\gamma}$ is the permutation symbol on $\{0, 1, 2\}$.

- Adjoint action

$$\hat{\mu}\,|\nu\rangle := |[\mu,\nu]\rangle \;\text{ for } \mu,\nu \in \mathfrak{se}(3) \tag{29}$$

$$\langle\phi_\alpha|\,\hat{\phi}_\beta\,|\phi_\gamma\rangle = c_{\alpha\beta\gamma} \tag{30}$$
$$\tag{31}$$

$$\langle\phi^\alpha|\,\hat{\phi}_\beta\,|\phi_\gamma\rangle = c^\alpha_{\beta\gamma} \tag{32}$$
$$\tag{33}$$

- Some useful formulas

$$\phi_{2\alpha}, \phi_{2\alpha+1} = \begin{bmatrix} \hat{e}_\alpha & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_\alpha \\ 0 & 0 \end{bmatrix} \tag{34}$$

$$\phi_{2\beta} = \sum_{\alpha,\gamma\in\mathbb{Z}_3} \varepsilon_{\alpha\beta\gamma}\phi_{2\alpha+1}\phi^T_{2\gamma+1} \text{ for } \beta \in \mathbb{Z}_3 \tag{35}$$

$$\phi^0, \phi^2, \phi^4 = \frac{1}{2}\phi_0, \frac{1}{2}\phi_2, \frac{1}{2}\phi_4 \tag{36}$$
$$\phi^1, \phi^3, \phi^5 = \phi_1, \phi_3, \phi_5 \tag{37}$$

$$\hat{\phi}_{2\alpha}\phi_{2\beta} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma} \tag{38}$$
$$\hat{\phi}_{2\alpha}\phi_{2\beta+1} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma+1} \tag{39}$$
$$\hat{\phi}_{2\alpha+1}\phi_{2\beta} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma+1} \tag{40}$$
$$\hat{\phi}_{2\alpha+1}\phi_{2\beta+1} = 0 \tag{41}$$

$$c^{2\alpha}_{2\beta,2\gamma} = c^{2\alpha+1}_{2\beta,2\gamma+1} = c^{2\alpha+1}_{2\beta+1,2\gamma} = \varepsilon_{\alpha\beta\gamma} \tag{42}$$
$$c^{2\alpha+1}_{2\beta+1,2\gamma+1} = 0 \tag{43}$$

$$
\begin{aligned}
\left(c^{\dagger}\right)^{\alpha}_{\beta\gamma} &= \langle \phi^{\alpha} | \, \hat{\phi}^{\dagger}_{\beta} \, | \phi_{\gamma} \rangle && (44)\\
&= \left\langle \hat{\phi}_{\beta} \phi^{\alpha} \middle| \phi_{\gamma} \right\rangle && (45)\\
&= \left\langle \phi_{\gamma} \middle| \hat{\phi}_{\beta} \phi^{\alpha} \right\rangle && (46)\\
&= g^{\alpha\mu} g_{\gamma\nu} \, \langle \phi^{\nu} | \, \hat{\phi}_{\beta} \, | \phi_{\mu} \rangle && (47)\\
&= \sum_{\mu,\nu=0}^{5} \frac{2}{3+(-1)^{\alpha}} \delta_{\alpha\mu} \frac{3+(-1)^{\gamma}}{2} \delta_{\gamma\nu} c^{\nu}_{\beta\mu} && (48)\\
&= \sum_{\mu,\nu=0}^{5} \frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} \delta_{\alpha\mu} \delta_{\gamma\nu} c^{\nu}_{\beta\mu} && (49)\\
&= \frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} c^{\gamma}_{\beta\alpha} \text{ (no sum over repeated indices)} && (50)
\end{aligned}
$$

$$
\frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} = \begin{cases} 1 & \text{if } \alpha+\gamma \text{ is even} \\ 2 & \text{if } \alpha \text{ is odd }, \gamma \text{ is even} \\ 1/2 & \text{if } \gamma \text{ is odd }, \alpha \text{ is even} \end{cases} \qquad (51)
$$

1.1.5 Matrix-valued forms

The basis (20-21) and inner product (22) induce a left invariant frame field and a left invariant metric on $SE(3)$

$$T_\rho SE(3) = \text{span}\{\rho\phi_\alpha\}_{\alpha=0}^{5},\ \forall \rho \in SE(3) \tag{52}$$

$$\langle \rho\phi_\alpha | \rho\phi_\beta \rangle := \langle \phi_\alpha | \phi_\beta \rangle_{\mathfrak{se}(3)} \tag{53}$$

Define $\mathcal{G}$ to be the $SE(3)$ bundle over S and define $\mathfrak{g}^*$ to be the $\mathfrak{se}(3)$ bundle over T^*S. Any smooth section of $\mathcal{G}$ is determined up to a constant $SE(3)$ matrix, by a unique section section of $\mathfrak{g}^*$,

$$\begin{aligned}
\rho :& S \to SE(3) && (54)\\
\psi :& S \to \mathfrak{se}(3) \otimes T^*S && (55)\\
\psi =& \rho^{-1}\,\mathrm{d}\rho && (56)\\
=& \psi_i\,\mathrm{d}s^i && (57)\\
=& \psi_i^\alpha \phi_\alpha\,\mathrm{d}s^i\,. && (58)
\end{aligned}$$

which satisfies the structure equation

$$0 = \mathrm{d}^2\rho = \rho(\mathrm{d}\psi + \psi \wedge \psi) \tag{59}$$

Or, in local coordinates

$$\partial_i\,|\psi_j\rangle - \partial_j\,|\psi_i\rangle + \hat{\psi}_i\,|\psi_j\rangle = 0 \tag{60}$$

We can introduce a metric $\langle\cdot|\cdot\rangle$ on $\mathfrak{g}^*$ by taking the tensor product of the left invariant metric on $SE(3)$ with the minkowski metric on S.

$$\left\langle \rho\phi_\alpha\,\mathrm{d}s^i \middle| \rho\phi_\alpha\,\mathrm{d}s^i \right\rangle := \langle \phi_\alpha | \phi_\beta \rangle_{\mathfrak{se}(3)} \left\langle \mathrm{d}s^i \middle| \mathrm{d}s^j \right\rangle_S \tag{61}$$

The bundle $\mathfrak{g}^*$, decomposes with respect to (61), into an orthogonal direct sum of space-like angular, time-like angular, space-like linear, and time-like

linear subspaces:

$$|\psi\rangle = \sum_{\alpha \text{ even}} \sum_{k=1}^{p} \psi_k^\alpha \left|\phi_\alpha \, \mathrm{d}s^k\right\rangle \qquad \text{(space-like angular)} \tag{62}$$

$$+ \sum_{\alpha \text{ even}} \psi_0^\alpha \left|\phi_\alpha \, \mathrm{d}s^0\right\rangle \qquad \text{(time-like angular)} \tag{63}$$

$$+ \sum_{\alpha \text{ odd}} \sum_{k=1}^{p} \psi_k^\alpha \left|\phi_\alpha \, \mathrm{d}s^k\right\rangle \qquad \text{(space-like linear)} \tag{64}$$

$$+ \sum_{\alpha \text{ odd}} \psi_0^\alpha \left|\phi_\alpha \, \mathrm{d}s^0\right\rangle \qquad \text{(time-like linear)} \tag{65}$$

1.2 Dynamics

1.2.1 $\mathfrak{se}(3)$ sigma-model action

Let E be a functional of the form

$$E[\rho] = \int_S \frac{1}{2} \langle \mathrm{d}\rho | \, B(\rho, s) \, | \mathrm{d}\rho \rangle \, \mathrm{d}^{p+1}s \tag{66}$$

where the tensor

$$B : T^*SE(3) \to T^*SE(3) \tag{67}$$

$$B(\rho) : T^*_\rho SE(3) \to T^*_\rho SE(3) \tag{68}$$

$$B(\rho) = \sum_{\alpha,\beta=0}^{5} \sum_{i,j=0}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle \rho\phi_\beta \, \mathrm{d}s^j\right| \tag{69}$$

describes inertia and stiffness of the medium. We require that B satisfy the following:

- B is self-adjoint with respect to the metric (61)

$$B = B^\dagger \tag{70}$$

- B is $SE(3)$-equivariant

$$\langle \gamma\mu | \, B(\gamma) \, | \gamma\nu \rangle = \langle \mu | \, B(1) \, | \nu \rangle \text{ for all } \gamma \in SE(3) \text{ and } \mu, \nu \in \mathfrak{se}(3) \tag{71}$$

- B preserves the four subspaces mentioned in (62-65), is positive definite when restricted to a timelike subspace, and negative definite when restricted to a spacelike subspace.

$$B = \sum_{\alpha,\beta \text{ even}} \sum_{i,j=1}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle\rho\phi_\beta \, \mathrm{d}s^j\right| \qquad \text{(Rotational stiffness)} \tag{72}$$

$$+ \sum_{\alpha,\beta \text{ even}} B_{00}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^0\right\rangle \left\langle\rho\phi_\beta \, \mathrm{d}s^0\right| \qquad \text{(Rotational intertia)} \tag{73}$$

$$+ \sum_{\alpha,\beta \text{ odd}} \sum_{i,j=1}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle\rho\phi_\beta\beta \, \mathrm{d}s^j\right| \qquad \text{(Linear stiffness)} \tag{74}$$

$$+ \sum_{\alpha,\beta \text{ odd}} B_{00}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^0\right\rangle \left\langle\rho\phi_\beta\beta \, \mathrm{d}s^0\right| \qquad \text{(Mass)} \tag{75}$$

If these conditions are satisfied by B, then the Lagrangian density

$$\mathcal{L} = \frac{1}{2} \left\langle \mathrm{d}\rho \right| B(\rho, s) \left| \mathrm{d}\rho \right\rangle \tag{76}$$

is invariant under the transformation

$$\rho \mapsto \gamma\rho \tag{77}$$

$$\mathrm{d}\rho \mapsto \gamma \, \mathrm{d}\rho \tag{78}$$

for any $\gamma \in SE(3)$, so, without loss of generality we can take $\mathcal{L}$ to be a function of $\psi = \rho^{-1} \, \mathrm{d}\rho$ and s.

$$\mathcal{L}(\rho, \mathrm{d}\rho \, , s) = \mathcal{L}(1, \rho^{-1} \, \mathrm{d}\rho \, , s) = \mathcal{L}(\psi, s) \tag{79}$$

1.2.2 Variational derivatives

Suppose $I \subset \mathbb{R}$ is a compact interval, and the fields

$$\rho : S \to SE(3) \tag{80}$$

$$\rho(s) = \begin{bmatrix} Q(s) & r(s) \\ 0 & 1 \end{bmatrix} \tag{81}$$

$$\tilde{\rho} : I \times S \to SE(3) \tag{82}$$

$$\tilde{\rho}(\xi, s) = \begin{bmatrix} \tilde{Q}(\xi, s) & \tilde{r}(\xi, s) \\ 0 & 1 \end{bmatrix} \tag{83}$$

satisfy

$$\tilde{\rho}(0, s) = \rho(s). \tag{84}$$

It follows that the corresponding $\mathfrak{se}(3)$-valued 1-forms

$$\psi = \rho^{-1}\,\mathrm{d}\rho \tag{85}$$

$$\psi(s) = \psi_i(s)\,\mathrm{d}s^i \tag{86}$$

$$\tilde{\psi} = \tilde{\rho}^{-1}\,\mathrm{d}\tilde{\rho} \tag{87}$$

$$\tilde{\psi}(\xi, s) = \tilde{\psi}_i(\xi, s)\,\mathrm{d}s^i + \tilde{\psi}_\xi(\xi, s)\,\mathrm{d}\xi \tag{88}$$

must satisfy

$$\tilde{\psi}(0, s) = \psi(s) + \tilde{\psi}_\xi(0, s)\,\mathrm{d}\xi \tag{89}$$

$$\tilde{\psi}_i(0, s) = \psi_i(s). \tag{90}$$

$$\partial_j \psi_i = \left(\partial_i + \hat{\psi}_i\right) \psi_j \tag{91}$$

$$\partial_\xi \tilde{\psi}_i(0, s) = \left(\partial_i + \hat{\psi}_i(s)\right) \tilde{\psi}_\xi(0, s). \tag{92}$$

In traditional calculus of variations notation, $\partial_\xi \tilde{\psi}_i(0, s)$ would be written as $\delta\psi_i(s)$ and (92) would read

$$\delta\psi_i = \left(\partial_i + \hat{\psi}_i\right) \psi_\xi. \tag{93}$$

The quantity $\tilde{\psi}_\xi$ replaces $\delta\psi$ as the fundamental field variation, and is used to construct covariant variational expressions.

1.2.3 The Euler-Poincare Equations

The first variation of E in the direction of $\delta\rho = \rho\chi$, where χ is any $\mathfrak{se}(3)$-valued field on S satisfying

$$\delta\psi_k = \left(\partial_k + \hat{\psi}_k\right)\chi, \tag{94}$$

is given by

$$\begin{aligned}
\delta E_\rho(\rho\chi) &= \left\langle \frac{\delta E}{\delta\psi_k} \middle| \delta\psi_k \right\rangle_{T\mathcal{G}} && (95)\\
&= \left\langle \frac{\delta E}{\delta\psi_k} \middle| \left(\partial_k + \hat{\psi}_k\right)\chi \right\rangle_{T\mathcal{G}} && (96)\\
&= \left\langle \left(\partial_k + \hat{\psi}_k\right)^\dagger \frac{\delta E}{\delta\psi_k} \middle| \chi \right\rangle_{T\mathcal{G}} && (97)\\
&= \int_S \partial_k \left\langle \frac{\delta E}{\delta\psi_k(s)} \middle| \chi(s) \right\rangle_{\mathfrak{se}(3)} + \left\langle \left(-\partial_k + \hat{\psi}_k(s)^\dagger\right) \frac{\delta E}{\delta\psi_k(s)} \middle| \chi(s) \right\rangle_{\mathfrak{se}(3)}, && (98)
\end{aligned}$$

so, the Euler-Poincare equations are

$$\left(-\partial_k + \hat{\psi}_k(s)^\dagger\right) \frac{\delta E}{\delta\psi_k(s)} = 0 \tag{99}$$

$$\left.\frac{\delta E}{\delta\psi_k(s)}\right|_{\partial S} = 0. \tag{100}$$

We can use the structure constants (28) to express (99-100) in component form,

$$-\partial_k \frac{\partial\mathcal{L}}{\partial\psi_k^\alpha} + c_{\gamma\alpha}^\beta \psi_k^\gamma \frac{\partial\mathcal{L}}{\partial\psi_k^\beta} = 0, \text{ for } 0 \le \alpha \le 5 \tag{101}$$

$$\left.\frac{\partial\mathcal{L}}{\partial\psi_k^\alpha}\right|_{\partial S} = 0. \tag{102}$$

For the Lagrangian (76), the extremal conditions (99) and (100) read

$$\partial_j \left(B^{jk}_{\alpha\beta} \psi^\beta_k \right) + c^\beta_{\alpha\gamma} B^{jk}_{\beta\mu} \psi^\gamma_j \psi^\mu_k = 0 \tag{103}$$

Or, separating inertial and elastic stresses,

$$\partial_t \left(B^{tt}_{\alpha\beta} \psi^\beta_t \right) + c^\beta_{\alpha\gamma} B^{tt}_{\beta\mu} \psi^\gamma_t \psi^\mu_t = - \sum_{j=1}^{p} \sum_{k=1}^{p} \partial_j \left(B^{jk}_{\alpha\beta} \psi^\beta_k \right) + c^\beta_{\alpha\gamma} B^{jk}_{\beta\mu} \psi^\gamma_j \psi^\mu_k \tag{104}$$

Any nonconservative N_α or constraint C_α forces would appear on the right-hand side,

$$\partial_t \left(B^{tt}_{\alpha\beta} \psi^\beta_t \right) + c^\beta_{\alpha\gamma} B^{tt}_{\beta\mu} \psi^\gamma_t \psi^\mu_t = - \sum_{j=1}^{p} \sum_{k=1}^{p} \left(\partial_j \left(B^{jk}_{\alpha\beta} \psi^\beta_k \right) + c^\beta_{\alpha\gamma} B^{jk}_{\beta\mu} \psi^\gamma_j \psi^\mu_k \right) + N_\alpha + C_\alpha \tag{105}$$

The even $\alpha = 0, 2, 4$ equations express conservation of angular momentum and the odd $\alpha = 1, 3, 5$ express the conservation of linear momentum.

1.3 The Kirchhoff Rod ($p = 1$)

We enforce inextensibility and unshearability constraints of the Kirchhoff rod,

$$\boldsymbol{q}_0 \cdot \partial_s \boldsymbol{r}, \boldsymbol{q}_1 \cdot \partial_s \boldsymbol{r}, \boldsymbol{q}_2 \cdot \partial_s \boldsymbol{r} = \psi^1_s, \psi^3_s, \psi^5_s = 1, 0, 0, \tag{106}$$

by adding an additional term to the Lagrangian (76),

$$\mathcal{L} = \frac{1}{2} \langle \psi | B \psi \rangle - \sum_{\mu \in \mathbb{Z}_3} \lambda_\mu \psi^{2\mu+1}_s \tag{107}$$

The constraints depend only on ψ, so, (101-102, 105) are still valid and the equations of motion are

$$\partial_t \left(B^{tt}_{\alpha\beta} \psi^\beta_t \right) + c^\beta_{\alpha\gamma} B^{tt}_{\beta\mu} \psi^\gamma_t \psi^\mu_t = - \partial_s \left(B^{ss}_{\alpha\beta} \psi^\beta_s \right) - c^\beta_{\alpha\gamma} B^{ss}_{\beta\mu} \psi^\gamma_s \psi^\mu_s + N_\alpha \tag{108}$$

$$+ \sum_{\mu \text{ odd}} \partial_s \lambda_\mu \delta^\mu_\alpha + c^\mu_{\alpha\gamma} \psi^\gamma_s \lambda_\mu \tag{109}$$

The Darboux tensor $\hat{\boldsymbol{\Omega}}$, angular velocity tensor $\hat{\boldsymbol{\omega}}$, unit tangent $\boldsymbol{e}_1$, and linear velocity vector $\boldsymbol{u}$,

$$\hat{\boldsymbol{\Omega}} = \begin{bmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{bmatrix}, \quad \hat{\boldsymbol{\omega}} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}, \quad \boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{110}$$

are related to ψ by

$$\sum_{\alpha \text{ even}} \psi_1^\alpha \phi_\alpha \, \mathrm{d}s^1 = \begin{bmatrix} \hat{\boldsymbol{\Omega}} & 0 \\ 0 & 0 \end{bmatrix} \mathrm{d}s \tag{111}$$

$$\sum_{\alpha \text{ even}} \psi_0^\alpha \phi_\alpha \, \mathrm{d}s^0 = \begin{bmatrix} \hat{\boldsymbol{\omega}} & 0 \\ 0 & 0 \end{bmatrix} \mathrm{d}t \tag{112}$$

$$\sum_{\alpha \text{ odd}} \psi_1^\alpha \phi_\alpha \, \mathrm{d}s^1 = \begin{bmatrix} 0 & \boldsymbol{e}_1 \\ 0 & 0 \end{bmatrix} \mathrm{d}s \tag{113}$$

$$\sum_{\alpha \text{ odd}} \psi_0^\alpha \phi_\alpha \, \mathrm{d}s^0 = \begin{bmatrix} 0 & \boldsymbol{u} \\ 0 & 0 \end{bmatrix} \mathrm{d}t \tag{114}$$

Suppose the nonconservative force N_α is the sum of a viscous drag term which is linear in the timelike components of ψ,

$$N_\alpha = -\eta_{\alpha\beta} \psi_t^\beta, \tag{115}$$

and an active stress term A_α. Consider the limit in which inertial forces are negligible

$$B_{\alpha\beta}^{tt} \to 0. \tag{116}$$

Our equations of motion are then

$$\boxed{\begin{gathered} -\partial_s \left(B_{\alpha\beta}^{ss} \psi_s^\beta \right) - c_{\alpha\gamma}^\beta B_{\beta\mu}^{ss} \psi_s^\gamma \psi_s^\mu + \sum_{\mu \text{ odd}} \left(\partial_s \lambda_\mu \delta_\alpha^\mu - c_{\gamma\alpha}^\mu \psi_s^\gamma \lambda_\mu \right) + A_\alpha - \eta_{\alpha\beta} \psi_t^\beta = 0 \\ \partial_t \psi_s^\alpha - \partial_s \psi_t^\alpha = c_{\beta\gamma}^\alpha \psi_s^\beta \psi_t^\gamma \\ \psi_s^1, \psi_s^3, \psi_s^5 = 1, 0, 0 \end{gathered}} \tag{117}$$

By equation (106), the centerline tangent is constant in the material frame

$$\partial_t \psi_1^1 = \partial_t \psi_1^3 = \partial_t \psi_1^5 = 0 \tag{118}$$

$$\partial_t \left(Q^T \partial_s \boldsymbol{r} \right) = 0 \tag{119}$$

and the odd $\alpha = 1, 3, 5$ components of the structure equation reduce to

$$\partial_s \psi_t^\alpha + c_{\beta\gamma}^\alpha \psi_s^\beta \psi_t^\gamma = 0 \text{ for } \alpha \text{ odd.} \tag{120}$$

Or in vector notation

$$\partial_s \left(Q^T \partial_t \boldsymbol{r} \right) + \boldsymbol{\Omega} \times \left(Q^T \partial_t \boldsymbol{r} \right) = 0 \tag{121}$$

1.3.1 Notation and some facts about Orthogonal matrices

Canonical basis for $\mathbb{R}^3$

$$\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{122}$$

Canonical basis for $\mathfrak{so}(3)$ (cross product matrices)

$$\hat{\boldsymbol{e}}_0, \hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{123}$$

For $\boldsymbol{u} = u_0 \boldsymbol{e}_0 + u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2$ define $\hat{\boldsymbol{u}} := u_0 \hat{\boldsymbol{e}}_0 + u_1 \hat{\boldsymbol{e}}_1 + u_2 \hat{\boldsymbol{e}}_2$. If P is a rotation then

$$\boldsymbol{w} = P\boldsymbol{u} \Leftrightarrow \hat{\boldsymbol{w}} = P \hat{\boldsymbol{u}} P^T \tag{124}$$

Rotations about coordinate axes

$$e^{\theta \hat{\boldsymbol{e}}_0}, e^{\theta \hat{\boldsymbol{e}}_1}, e^{\theta \hat{\boldsymbol{e}}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}, \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{pmatrix}, \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{125}$$

If $Q \in SO(3)$ transforms $\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2$ to $\boldsymbol{q}_0 = Q\boldsymbol{e}_0, \boldsymbol{q}_1 = Q\boldsymbol{e}_1, \boldsymbol{q}_2 = Q\boldsymbol{e}_2$ then rotations about $\boldsymbol{q}_i$'s are given by

$$e^{\theta \hat{\boldsymbol{q}}_i} = Q e^{\theta \hat{\boldsymbol{e}}_i} Q^{-1} \tag{126}$$

1.4 Framed curves

A curve $\boldsymbol{r}$ with adapted frame $\boldsymbol{q}_0 = \boldsymbol{r}', \boldsymbol{q}_1, \boldsymbol{q}_2$ can be used to define an $SE(3)$-valued matrix

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{q}_0\boldsymbol{e}_0^T + \boldsymbol{q}_1\boldsymbol{e}_1^T + \boldsymbol{q}_2\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{127}$$

The *shape* of the framed curve, or characteristics invariant under rigid transformation, can be described by a curve through the special Euclidean algebra $\mathfrak{se}(3)$

$$\psi = \rho^{-1}\rho' = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \tag{128}$$

Requiring $\boldsymbol{q}_0 = \boldsymbol{r}'$ is equivalent to $\boldsymbol{U} = \boldsymbol{e}_0$. The framed curve can be recovered from ψ by solving the initial value problem

$$\rho' = \rho\psi, \;\; \rho(0) = \rho_0 \tag{129}$$

The solution to (129)

$$\rho(s) = \rho_0 Oexp(\psi, s) \tag{130}$$

is given in terms of the ordered exponential

$$Oexp(\psi, s) = \prod_{\xi=0}^{\xi=s} e^{\psi(\xi)\mathrm{d}\xi} = \lim_{n\to\infty} e^{\psi(\xi_0)\mathrm{d}\xi} e^{\psi(\xi_1)\mathrm{d}\xi} \ldots e^{\psi(\xi_{n-1})\mathrm{d}\xi} \tag{131}$$

where $0 = \xi_0 < \xi_1 < \ldots < \xi_{n-1} = s$ is a partition of the interval $[0, s]$. The related problem

$$\rho' = \underline{\psi}\rho \tag{132}$$

$$\rho(0) = \rho_0, \tag{133}$$

where $\underline{\psi} = \rho\psi\rho^{-1}$, can be solved by reversing the above path ordering

$$\rho(s) = \lim_{n\to\infty} e^{\underline{\psi}(\xi_{n-1})\mathrm{d}\xi} e^{\underline{\psi}(\xi_{n-2})\mathrm{d}\xi} \ldots e^{\underline{\psi}(\xi_0)\mathrm{d}\xi} \rho_0 \tag{134}$$

Suppose $\boldsymbol{r}$ has Frenet-Serret frame $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ curvature κ and torsion τ.

$$\mu = \begin{pmatrix} F & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{135}$$

$$\mu^{-1}\mu' = \begin{pmatrix} \tau\hat{\boldsymbol{e}}_0 + \kappa\hat{\boldsymbol{e}}_2 & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \tag{136}$$

Any two adapted frames are related by rotation about curve tangent. As a consequence, we find the following

$$Q = e^{-\phi\hat{\boldsymbol{T}}}F = Fe^{-\phi\hat{\boldsymbol{e}}_0} \tag{137}$$

$$Q\boldsymbol{U} = F\boldsymbol{U} = \boldsymbol{r}' \tag{138}$$

$$\phi(s) = \phi_0 + \int_0^s \tau - \Omega_0 \,\mathrm{d}s \tag{139}$$

$$\boldsymbol{\Omega} = \Omega_0\boldsymbol{e}_0 + \boldsymbol{\Omega}_\perp = \Omega_0\boldsymbol{e}_0 + \Omega_1\boldsymbol{e}_1 + \Omega_2\boldsymbol{e}_2 \tag{140}$$

$$\boldsymbol{\Omega}_\perp = \kappa e^{\phi\hat{\boldsymbol{e}}_0}\boldsymbol{e}_2 \tag{141}$$

$$Q\boldsymbol{\Omega}_\perp = \boldsymbol{r}' \times \boldsymbol{r}'' = \kappa\boldsymbol{B} \tag{142}$$

$$\kappa = |\boldsymbol{\Omega}_\perp| \tag{143}$$

$$\tau = \Omega_0 + \frac{\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}'_\perp)}{|\boldsymbol{\Omega}_\perp|^2} = \Omega_0 + \phi' \tag{144}$$

1.5 Time dependent framed curves

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix}, \ \psi = \rho^{-1}\,\mathrm{d}\rho \tag{145}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix}\mathrm{d}s + \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix}\mathrm{d}t \tag{146}$$

$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{147}$$

Angular $\boldsymbol{\omega}$ and linear $\boldsymbol{u}$ velocities appear naturally as timelike analogues of the twist/curvature $\boldsymbol{\Omega}$ and curve tangent $\boldsymbol{U}$. Commutativity of partial derivatives requires ψ satisfy an additional *structure equation*

$$\rho^{-1}\,\mathrm{d}^2\rho = \mathrm{d}\psi + \psi \wedge \psi = 0. \tag{148}$$

The structure equation (148) together with the constraint (147) imply

$$\dot{\Omega} = \boldsymbol{\omega}' + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{149}$$

$$\boldsymbol{\omega} \times \boldsymbol{e}_0 = \boldsymbol{u}' + \boldsymbol{\Omega} \times \boldsymbol{u} \tag{150}$$

Note that imply the familiar consitency conditions found in

$$\dot{\Omega}_0 = \omega_0' + \boldsymbol{r}' \cdot (\boldsymbol{r}'' \times \dot{\boldsymbol{r}}') \tag{151}$$

$$\boldsymbol{r}' \cdot \dot{\boldsymbol{r}}' = 0 \tag{152}$$

Components of ψ are related to inertial frame derivatives of $\boldsymbol{r}$ by

$$\boldsymbol{\Omega} = \Omega_0 \boldsymbol{e}_0 + \Omega_1 \boldsymbol{e}_1 + \Omega_2 \boldsymbol{e}_2 \tag{153}$$

$$Q\boldsymbol{\Omega}_\perp = \boldsymbol{r}' \times \boldsymbol{r}'' = \kappa \boldsymbol{B} \tag{154}$$

$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{155}$$

$$\boldsymbol{\omega} = \omega_0 \boldsymbol{e}_0 + \omega_1 \boldsymbol{e}_1 + \omega_2 \boldsymbol{e}_2 \tag{156}$$

$$Q\boldsymbol{\omega}_\perp = \boldsymbol{r}' \times \dot{\boldsymbol{r}}' \tag{157}$$

$$\boldsymbol{u} = u_0 \boldsymbol{e}_0 + u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2 \tag{158}$$

$$Q\boldsymbol{u} = \dot{\boldsymbol{r}} \tag{159}$$

1.5.1 $U(1)$ gauge transformations

When representing a filament with circular cross-section by a framed curve, the initial direction chosen for the transverse frame vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$ is equivalent to fixing a gauge. The gauge group $U(1)$ acts by rotating the frame at each point along the curve by a constant angle about the curve tangent $\boldsymbol{q}_0 = \boldsymbol{r}'$. Gauge invariants include the centerline $\boldsymbol{r}$ and all associated geometric quatities (e.g. curvature, torsion, and Frenet-Serret frame), as well as the twist Ω_0.

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix}, \ \psi = \rho^{-1}\,\mathrm{d}\rho \tag{160}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \mathrm{d}s + \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix} \mathrm{d}t \tag{161}$$

Gauge transformations can be represented by $SE(3)$-valued matrices

$$\gamma = \begin{pmatrix} e^{\theta \hat{\boldsymbol{e}}_0} & 0 \\ 0 & 1 \end{pmatrix} \tag{162}$$

whose action on ρ is defined by

$$\begin{aligned} \rho \mapsto \rho\gamma^{-1} &= \begin{pmatrix} Qe^{-\theta \hat{\boldsymbol{e}}_0} & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{-\theta \hat{\boldsymbol{q}}_0} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \end{aligned} \tag{163}$$

Under a change of guage, ψ transforms as

$$\psi \mapsto \gamma\psi\gamma^{-1}. \tag{164}$$

Or, in terms of its' components

$$\boldsymbol{\Omega} \mapsto \Omega_0 \boldsymbol{e}_0 + e^{\theta \hat{\boldsymbol{e}}_0} \boldsymbol{\Omega}_\perp \tag{165}$$

$$\boldsymbol{U} \mapsto \boldsymbol{U} \tag{166}$$

$$\boldsymbol{\omega} \mapsto \omega_0 \boldsymbol{e}_0 + e^{\theta \hat{\boldsymbol{e}}_0} \boldsymbol{\omega}_\perp \tag{167}$$

$$\boldsymbol{u} \mapsto u_0 \boldsymbol{e}_0 + e^{\theta \hat{\boldsymbol{e}}_0} \boldsymbol{u}_\perp \tag{168}$$

1.5.2 Symmetries

Let ρ describe an arclength parameterized $\boldsymbol{r}(s)$ and its' Frenet-Serret frame $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ with curvature κ and torsion τ.

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{169}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau \hat{\boldsymbol{e}}_0 + \kappa \hat{\boldsymbol{e}}_2 & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} \tag{170}$$

The curvature κ and torsion τ are even functions of s iff the curve is invariant under rotation by π about an axis which intersects the curve at $\boldsymbol{r}(0)$ and which is perpendicular to $\boldsymbol{T}(0)$. If the curvature is nonvanishing at $s = 0$,

then the axis of rotation is parallel to $\boldsymbol{n} = \boldsymbol{N}(0)$.

$$\psi(-s) = \psi(s) \tag{171}$$

$$\Leftrightarrow \tag{172}$$

$$\rho(-s) = \rho_0 \gamma \rho_0^{-1} \rho(s) \gamma^{-1} \tag{173}$$

$$= \begin{pmatrix} e^{\pi \hat{\boldsymbol{N}}(0)} Q(s) e^{\pi \hat{\boldsymbol{e}}_1} & \boldsymbol{r}(0) + e^{\pi \hat{\boldsymbol{N}}(0)} \left(\boldsymbol{r}(s) - \boldsymbol{r}(0) \right) \\ 0 & 1 \end{pmatrix} \tag{174}$$

where γ is given by

$$\gamma = \begin{pmatrix} e^{\pi \hat{\boldsymbol{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} \tag{175}$$

Proof.

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{176}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau \hat{\boldsymbol{e}}_0 + \kappa \hat{\boldsymbol{e}}_2 & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} \tag{177}$$

To see this, first, assume $\rho_0 = 1$, so that $\boldsymbol{T}(0) = \boldsymbol{e}_0$, $\boldsymbol{N}(0) = \boldsymbol{e}_1$, $\boldsymbol{B}(0) = \boldsymbol{e}_2$ and $\boldsymbol{r}(0) = 0$.

$$\rho(s) = \prod_{\xi=0}^{\xi=s} e^{\psi(\xi) \mathrm{d}\xi} \tag{178}$$

$$\rho(-s) = \prod_{\xi=0}^{\xi=-s} e^{\psi(\xi) \mathrm{d}\xi} = \prod_{\xi=0}^{\xi=s} e^{-\psi(-\xi) \mathrm{d}\xi} = \prod_{\xi=0}^{\xi=s} e^{-\psi(\xi) \mathrm{d}\xi} \tag{179}$$

Where the last equality uses $\psi(-s) = \psi(s)$. If we define

$$\gamma = \begin{pmatrix} e^{\pi \hat{\boldsymbol{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\boldsymbol{e}_0 \boldsymbol{e}_0^T + \boldsymbol{e}_1 \boldsymbol{e}_1^T - \boldsymbol{e}_2 \boldsymbol{e}_2^T & 0 \\ 0 & 1 \end{pmatrix} \tag{180}$$

then we have

$$\gamma \psi \gamma^{-1} = \begin{pmatrix} e^{\pi \hat{\boldsymbol{e}}_1} \left(\tau \hat{\boldsymbol{e}}_0 + \kappa \hat{\boldsymbol{e}}_2 \right) e^{-\pi \hat{\boldsymbol{e}}_1} & e^{\pi \hat{\boldsymbol{e}}_1} \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -\tau \hat{\boldsymbol{e}}_0 - \kappa \hat{\boldsymbol{e}}_2 & -\boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = -\psi \tag{181}$$

$$\rho(-s) = \prod_{\xi=0}^{\xi=s} e^{\gamma\psi(\xi)\gamma^{-1}\mathrm{d}\xi} = \gamma \left(\prod_{\xi=0}^{\xi=s} e^{\psi(\xi)\mathrm{d}\xi} \right) \gamma^{-1} = \gamma\rho(s)\gamma^{-1} \tag{182}$$

$$\gamma\rho\gamma^{-1} = \begin{pmatrix} e^{\pi\hat{\boldsymbol{e}}_1} Q e^{-\pi\hat{\boldsymbol{e}}_1} & e^{\pi\hat{\boldsymbol{e}}_1}\boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{183}$$

More generally, if $\rho(s) = \rho_0 Oexp(\psi, s)$, then $\rho_0^{-1}\rho$ satisfies $\rho_0^{-1}\rho(-s) = \gamma\rho_0^{-1}\rho(s)\gamma^{-1}$

$$\rho(-s)\rho_0\gamma\rho_0^{-1}\rho\gamma^{-1} = \rho(0) \begin{pmatrix} e^{\pi\hat{\boldsymbol{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} \rho(0)^{-1}\rho(s) \begin{pmatrix} e^{-\pi\hat{\boldsymbol{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} \tag{184}$$

□

1.5.3 Evolution equations – inertial frame

If Ω_0, κ, τ are even, then there exists a gauge in which Ω_1 is odd and Ω_2 is even. If m and f are also odd at $t = 0$, then this symmetry is conserved. Boundary conditions at $s = 0$ are given by

$$\Omega_0' = 0 \tag{185}$$
$$\Omega_1 = 0, \Omega_1'' = 0 \tag{186}$$
$$\Omega_2' = 0, \Omega_2''' = 0 \tag{187}$$
$$\lambda' = 0 \tag{188}$$

$$Q\boldsymbol{\Omega} = \Omega_0\boldsymbol{r}' + \boldsymbol{r}' \times \boldsymbol{r}'' \tag{189}$$
$$Q\boldsymbol{\omega} = \omega_0\boldsymbol{r}' + \boldsymbol{r}' \times \dot{\boldsymbol{r}}' \tag{190}$$
$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{191}$$
$$Q\boldsymbol{u} = \dot{\boldsymbol{r}} \tag{192}$$

Structure equation

$$\dot{\Omega}_0 = \omega_0' + \boldsymbol{r}' \cdot (\boldsymbol{r}'' \times \dot{\boldsymbol{r}}') \tag{193}$$
$$\boldsymbol{r}' \cdot \dot{\boldsymbol{r}}' = 0 \tag{194}$$

Force and moment balance

$$\left(\beta_\parallel \Omega_0 \boldsymbol{r}' + \beta_\perp \boldsymbol{r}' \times \boldsymbol{r}''\right)' + \boldsymbol{r}' \times \boldsymbol{\Lambda} + m\boldsymbol{r}' - \zeta_r \omega_0 \boldsymbol{r}' = 0 \tag{195}$$

$$\boldsymbol{\Lambda}' + f\boldsymbol{r}' - \zeta \dot{\boldsymbol{r}} = 0 \tag{196}$$

where $\zeta = \zeta_\parallel \boldsymbol{r}'\boldsymbol{r}'^T + \zeta_\perp(1 - \boldsymbol{r}'\boldsymbol{r}'^T)$. ... ...

Moment balance gives

$$\omega_0 = \frac{\beta_\parallel}{\zeta_r}\Omega_0' + \frac{1}{\zeta_r}m \tag{197}$$

$$\boldsymbol{\Lambda} = \lambda \boldsymbol{r}' + \boldsymbol{r}' \times \left(\beta_\parallel \Omega_0 \boldsymbol{r}'' + \beta_\perp \boldsymbol{r}' \times \boldsymbol{r}'''\right) \tag{198}$$

Dotting force balance with $\boldsymbol{r}'$ and taking a derivative wrt s and using $\boldsymbol{r}'\dot{\boldsymbol{r}}' = 0$ gives

$$\boldsymbol{r}' \cdot \boldsymbol{\Lambda}' + f - \zeta_\parallel \boldsymbol{r}' \cdot \dot{\boldsymbol{r}} = 0 \tag{199}$$

$$(\boldsymbol{r}' \cdot \boldsymbol{\Lambda}')' + f' - \zeta_\parallel \boldsymbol{r}'' \cdot \dot{\boldsymbol{r}} = 0 \tag{200}$$

Dotting force balance with $\boldsymbol{r}''$, solving for $\boldsymbol{r}'' \cdot \dot{\boldsymbol{r}}$, and using the above gives the tension equation

$$\boldsymbol{r}'' \cdot \dot{\boldsymbol{r}} = \frac{1}{\zeta_\perp}\boldsymbol{r}'' \cdot \boldsymbol{\Lambda}' \tag{201}$$

$$(\boldsymbol{r}' \cdot \boldsymbol{\Lambda}')' + f' - \frac{\zeta_\parallel}{\zeta_\perp}\boldsymbol{r}'' \cdot \boldsymbol{\Lambda}' = 0 \tag{202}$$

$$\lambda'' - \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\lambda + f' + \frac{1}{2}\beta_\perp(\kappa^2)'' + \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\tau\left(\beta_\parallel\Omega_0 - \beta_\perp\tau\right) + \frac{\zeta_\parallel\beta_\perp}{\zeta_\perp}\kappa\kappa'' = 0 \tag{203}$$

Torsion can be eliminated using

$$\Omega_0\kappa^2\tau = \kappa^2\Omega_0^2 + \Omega_0\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp') \tag{204}$$

$$\kappa\kappa'' - \kappa^2\tau^2 = \boldsymbol{\Omega}_\perp \cdot \boldsymbol{\Omega}_\perp'' - \kappa^2\Omega_0^2 - 2\Omega_0\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp') \tag{205}$$

$$\lambda'' - \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\lambda + f' + \frac{1}{2}\beta_\perp(\kappa^2)'' \tag{206}$$

$$+\frac{\zeta_\parallel\beta_\parallel}{\zeta_\perp}\left(\kappa^2\Omega_0^2 + \Omega_0\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp')\right) \tag{207}$$

$$+\frac{\zeta_\parallel\beta_\perp}{\zeta_\perp}\left(\boldsymbol{\Omega}_\perp \cdot \boldsymbol{\Omega}_\perp'' - \kappa^2\Omega_0^2 - 2\Omega_0\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp')\right) \tag{208}$$

$$= 0 \tag{209}$$

2 Swimming P. Mirabilis

2.1 mathematical model

... The cell is modeled as a semiflexible filament with active force and moment distributions. The standard

The cell is modeled as a semi-flexible filament (Kirchhoff rod) with active force $\boldsymbol{f}_a = f_a(s,t)\boldsymbol{e}_0$ and moment $\boldsymbol{m}_a = m_a(s,t)\boldsymbol{e}_0$ distributions, where $\boldsymbol{e}_0$ is the local tangent vector, s is the arc-length, and t is time. Viscous stresses are treated using resistive-force theory $\boldsymbol{f}_v = -c_\parallel \boldsymbol{u}_\parallel - c_\perp \boldsymbol{u}_\perp$ with an additional viscous moment $\boldsymbol{m}_v = -c_r \boldsymbol{\omega}_\parallel$ proportional to the tangential angular velocity $\boldsymbol{\omega}_\parallel$.

$$\delta \int \frac{1}{2} B_\parallel |\Omega_\parallel|^2 + \frac{1}{2} B_\perp |\Omega_\perp|^2 + \boldsymbol{\lambda} \cdot (\boldsymbol{Q}^{-1} \partial_s \boldsymbol{r} - \boldsymbol{e}_0)\, \mathrm{d}s = \delta W_a + \delta W_v.$$

Force and moment balance together with the commutation relation for $SE(3)$ result in a system of nonlinear PDEs which describe the evolution of the cell:

$$\partial_t \boldsymbol{\Omega}_\parallel - \left[\frac{\beta_\parallel}{\eta_r} + \frac{\beta_\parallel - \beta_\perp}{\eta_\perp} |\boldsymbol{\Omega}_\perp|^2 \right] \partial_s^2 \boldsymbol{\Omega}_\parallel = G_\parallel(\boldsymbol{\Omega}_\parallel, \partial_s \boldsymbol{\Omega}_\parallel, \boldsymbol{\Omega}_\perp \ldots, \partial_s^3 \boldsymbol{\Omega}_\perp, \lambda, \partial_s m_a),$$

$$\partial_t \boldsymbol{\Omega}_\perp + \frac{\beta_\perp}{\eta_\perp} \partial_s^4 \boldsymbol{\Omega}_\perp = G_\perp(\boldsymbol{\Omega}_\parallel, \ldots, \partial_s^2 \boldsymbol{\Omega}_\parallel, \boldsymbol{\Omega}_\perp \ldots, \partial_s^3 \boldsymbol{\Omega}_\perp, \lambda, \ldots, \partial_s^2 \lambda, f_a, \partial_s f_a, m_a),$$

$$\partial_s^2 \lambda - \frac{1}{\eta_\perp} |\boldsymbol{\Omega}_\perp|^2 \lambda = G_\lambda(\boldsymbol{\Omega}_\parallel, \boldsymbol{\Omega}_\perp \ldots, \partial_s^2 \boldsymbol{\Omega}_\perp, \partial_s f_a).$$

The system is governed by four dimensionless parameters:

$$\beta_\parallel = B_\parallel / (f_a^* L^3), \beta_\perp = B_\perp / (f_a^* L^3), \eta_\parallel = c_\perp / c_\parallel, \eta_r = c_r / (c_\parallel L^2)$$

2.2 Notation, force/moment balance, and constraints

Unless otherwise stated, all vector/tensors are represented by component matrices in the material frame.

- Canonical basis for $\mathbb{R}^3$

$$\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{210}$$

- Material frame — $Q \in SO(3)$

$$Q = Q_{ij}\boldsymbol{e}_i\boldsymbol{e}_j^T = \boldsymbol{q}_j\boldsymbol{e}_j^T \tag{211}$$

- Centerline position vector — $\boldsymbol{x} \in \mathbb{R}^3$

$$\boldsymbol{x} = x_i\boldsymbol{e}_i \tag{212}$$

- Inextensibility/unshearability constraint

$$\partial_s\left(Q\boldsymbol{x}\right) = Q\boldsymbol{e}_0 \tag{213}$$

- Cross-section pose $\rho \in SE(3)$

$$\rho = \begin{pmatrix} Q & Q\boldsymbol{x} \\ 0 & 1 \end{pmatrix} \tag{214}$$

- Darboux vector — $\boldsymbol{\Omega} = \Omega_i\boldsymbol{e}_i$

$$\partial_s Q\boldsymbol{e}_k = Q(\boldsymbol{\Omega} \times \boldsymbol{e}_k) \tag{215}$$

  - Twist — Ω_0
  - Curvature — Ω_1, Ω_2

- Translational velocity — $\boldsymbol{u} = u_i\boldsymbol{e}_i = Q^T\partial_t\left(Q\boldsymbol{x}\right)$
- Angular velocity — $\boldsymbol{\omega} = \omega_i\boldsymbol{e}_i$

$$\partial_t Q\boldsymbol{e}_k = Q(\boldsymbol{\omega} \times \boldsymbol{e}_k) \tag{216}$$

- s,t-generators, Pfaffian constraints

$$\psi_s = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix}, \qquad \psi_t = \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix} \tag{217}$$

$$\partial_s\rho = \rho\psi_s, \qquad \partial_t\rho = \rho\psi_t \tag{218}$$

Note that these constraints are equivalent to

$$\partial_s Q = Q\hat{\boldsymbol{\Omega}}, \qquad \partial_t Q = Q\hat{\boldsymbol{\omega}} \tag{219}$$

$$\partial_s(Q\boldsymbol{x}) = Q\boldsymbol{e}_1, \qquad \partial_t(Q\boldsymbol{x}) = Q\boldsymbol{u} \tag{220}$$

- Compatibility conditions

$$\partial_t \boldsymbol{\Omega} = \partial_s \boldsymbol{\omega} + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{221}$$

$$\partial_s \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{u} = \boldsymbol{\omega} \times \boldsymbol{e}_0 \tag{222}$$

- Internal force on s from $(s, L]$ section — $\boldsymbol{f} = f_i \boldsymbol{e}_i = \lambda \boldsymbol{e}_0 + f_1 \boldsymbol{e}_1 + f_2 \boldsymbol{e}_2$

- Stiffness tensor and internal moment on s from $(s, L]$ section — $B = B_\parallel \boldsymbol{e}_1 \boldsymbol{e}_1^T + B_\perp \left(\boldsymbol{e}_2 \boldsymbol{e}_2^T + \boldsymbol{e}_3 \boldsymbol{e}_3^T \right)$

$$\boldsymbol{m} = B \boldsymbol{\Omega} \tag{223}$$

$$\boldsymbol{f}_v = -c \boldsymbol{u} \tag{224}$$

- Axial drag and moment per length — $c_r \in \mathbb{R}$

$$\boldsymbol{m}_v = -c_r \omega_1 \boldsymbol{e}_1 \tag{225}$$

- Active force and moment per length — f_a, m_a

$$\boldsymbol{f}_a = f_a \boldsymbol{e}_1, \qquad \boldsymbol{m}_a = m_a \boldsymbol{e}_1 = \chi f_a \boldsymbol{e}_1 \tag{226}$$

- Trajectories of the body are governed by the field equations (117) which express moment balance ($\alpha = 0, 2, 4$), force balance ($\alpha = 1, 3, 5$), compatibility conditions, and inextensibility/unshearability. They are equivalent to the vector equations

$$\begin{aligned} B \partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B \boldsymbol{\Omega} + \boldsymbol{e}_1 \times \boldsymbol{f} - c_r \omega_1 \boldsymbol{e}_1 + m_a \boldsymbol{e}_1 &= 0 \\ \partial_s \boldsymbol{f} + \boldsymbol{\Omega} \times \boldsymbol{f} - c_\parallel \boldsymbol{u}_\parallel - c_\perp \boldsymbol{u}_\perp + f_a \boldsymbol{e}_1 &= 0 \\ \partial_s \partial_t Q &= \partial_t \partial_s Q \\ \partial_s \partial_t (Q \boldsymbol{x}) &= \partial_t \partial_s (Q \boldsymbol{x}) \\ \partial_s (Q \boldsymbol{x}) &= Q \boldsymbol{e}_1 \end{aligned} \tag{227}$$

If we neglect the active force and moment, then (227) are the standard equations used to describe motion of a Kirchhoff rod in a viscous fluid [1], [2].

2.3 Units and nondimensional parameters

Let the body length L, characteristic swimming speed U, and parallel drag coefficient $c_\parallel$ define our fundamental system of units.

- Body dimensions – $[L] = [\ell] = \text{ m}$
  - $L \sim 10^{-5}$ m
  - Body diameter – $\sim 10^{-6}$ m
- Characteristic swimming speed – $[U] = \frac{\text{m}}{\text{s}}$
  - $U \approx 1.5 \times 10^{-5} \frac{\text{m}}{\text{s}}$
- Elastic stresses
  - Twisting/bending stiffness – $[B_\parallel] = [B_\perp] = \text{ N} \cdot \text{m}^2$
    - $B_\parallel \approx$???
    - $B_\perp \approx 5 \times 10^{-22} - 1.4 \times 10^{-20}$ N $\cdot$ m^2
  - Elastic moment – $[\boldsymbol{m}] = \text{ N} \cdot \text{m}$
    - $\boldsymbol{m} = B_\parallel \boldsymbol{\Omega}_\parallel + B_\perp \boldsymbol{\Omega}_\perp$
  - Elastic force – $[\boldsymbol{f}_\perp] = \text{ N}$
    - $\boldsymbol{f}_\perp = B_\perp \boldsymbol{e}_1 \times \boldsymbol{\Omega}_\parallel + (B_\parallel - B_\perp)\Omega_1 \boldsymbol{\Omega}_\perp$
- Viscous Stresses
  - Translational drag coefficients – $[c_\parallel] = [c_\perp] = \frac{\text{N}\cdot\text{s}}{\text{m}^2}$
    - $c_\parallel \approx 4.95 \times 10^{-2} \frac{\text{N}\cdot\text{s}}{\text{m}^2}$
    - $c_\perp \approx 7.83 \times 10^{-2} \frac{\text{N}\cdot\text{s}}{\text{m}^2}$
  - Rotational drag coefficient – $[c_r] = \text{N} \cdot \text{s}$
    - $c_r \approx 1.25 \times 10^{-14}$ N $\cdot$ s
- Active stresses
  - Characteristic force density – $[f_a^*] = \frac{\text{N}}{\text{m}}$
    - $f_a^* \approx c_\parallel U = 7.43 \times 10^{-7}$

- Characteristic moment density – $[m_a^*] =$ N
  - $m_a^* \approx 0$
- Diffusion coefficient – $[\sigma] = \frac{\mathrm{m}^2}{s}$
  - $\sigma \approx UL \times 10^{-2} \approx 1.5^{-12} \frac{\mathrm{m}^2}{s}$

Nondimensional stuff:

- Drag ratios

$$\eta_\perp = \frac{c_\perp}{c_\parallel} \approx 1.585 \tag{228}$$

$$\eta_r = \frac{c_r}{c_\parallel L^2} \approx 2.53 \times 10^{-3} \tag{229}$$

- Elastic-active ratios

$$\beta_\parallel = \frac{B_\parallel}{f_a^* L^3} \approx ??? \tag{230}$$

$$\beta_\perp = \frac{B_\perp}{f_a^* L^3} \approx .673 - 18.8 \tag{231}$$

2.4 Twist/curvature evolution and tension

The field equations (117) can be reduced to a system of four coupled equations in the twist Ω_1, curvatures Ω_2, Ω_3, and tension λ

$$\partial_t \Omega_1 + \left[\frac{B_\perp - B_\parallel}{c_\perp}|\boldsymbol{\Omega}_\perp|^2 - \frac{B_\parallel}{c_r}\right]\partial_s^2 \Omega_1 = G_1\left(\Omega_1, \partial_s\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \partial_s m_a\right) \tag{232}$$

$$\partial_t \boldsymbol{\Omega}_\perp + \frac{B_\perp}{c_\perp}\partial_s^4\boldsymbol{\Omega}_\perp + \frac{B_\perp - B_\parallel}{c_\perp}\boldsymbol{e}_1 \times \boldsymbol{\Omega}_\perp \partial_s^3 \Omega_1 = G_\perp\left(\Omega_1, \ldots, \partial_s^2\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \ldots, \partial_s^2\lambda, f_a, \partial_s f_a, m_a\right) \tag{233}$$

$$\partial_s^2 \lambda - \frac{c_\parallel}{c_\perp}|\boldsymbol{\Omega}_\perp|^2\lambda = G_\lambda\left(\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^2\boldsymbol{\Omega}_\perp, \partial_s f_a\right) \tag{234}$$

To show this, we need to express $f_2, f_3, \omega_1, \omega_2, \omega_3, u_1, u_2, u_3$ in terms of $\Omega_1, \Omega_2, \Omega_3, \lambda$. From force/moment balance (227) and the translational part

of the structure equation (222), we find

$$\boldsymbol{f} = \lambda \boldsymbol{e}_1 + \boldsymbol{e}_1 \times (B\partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B\boldsymbol{\Omega}) \tag{235}$$

$$\boldsymbol{u} = c^{-1} (\partial_s \boldsymbol{f} + \boldsymbol{\Omega} \times \boldsymbol{f} + f_a \boldsymbol{e}_1) \tag{236}$$

$$\boldsymbol{\omega} = \frac{1}{c_r} [\boldsymbol{e}_1 \cdot (B\partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B\boldsymbol{\Omega}) + m_a] \boldsymbol{e}_1 + \boldsymbol{e}_1 \times (\partial_s \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{u}) . \tag{237}$$

Substituting (235) into (236) and then the result into (237) eliminates the desired variables. Finally, the tension ODE (234) comes from $\boldsymbol{e}_1$ component of (222) and the curvature PDE's (232-233) come from (221).

2.5 Active force/moment evolution

$$\partial_t f_a = \frac{f_a^*}{L} \left[1 - \left(\frac{f_a}{f_a^*} \right)^2 \right] u_1 + \sigma \partial_s^2 f_a \tag{238}$$

$$\partial_s f_a |_{s=0,L} = 0 \tag{239}$$

$$m_a = \frac{m_a^*}{f_a^*} f_a \tag{240}$$

The active power per unit length is

$$\partial_s P_{active} = f_a u_1 + m_a \omega_1 \tag{241}$$

$$= \frac{B_\perp}{c_\parallel} \boldsymbol{\Omega}_\perp \cdot \partial_s \boldsymbol{\Omega}_\perp f_a + \frac{1}{c_\parallel} \partial_s \lambda f_a + \frac{1}{c_\parallel} f_a^2 + \frac{B_\parallel}{c_r} \partial_s \Omega_1 m_a + \frac{1}{c_r} m_a^2 \tag{242}$$

$$= \partial_s \left(\frac{B_\perp}{2\mu_\parallel} |\boldsymbol{\Omega}_\perp|^2 + \frac{1}{c_\parallel} \lambda + \frac{m_a^* B_\parallel}{f_a^* c_r} \Omega_1 \right) f_a + \left(\frac{1}{c_\parallel} + \frac{m_a^{*2}}{f_a^{*2} c_r} \right) f_a^2 \tag{243}$$

2.6 Pose integration

To integrate the frame and centerline we need the curvatures and twist for all $s \in [0, L]$, and the pose of a distinguished cross-section. Let's take the distinguished cross-section to be at $s = s_0$. The evolution equation for the cross-section is

$$\partial_t \rho_0 = \rho_0 \psi_0 \tag{244}$$

where

$$\rho_0 = \rho|_{s=s_0}\,, \qquad \boldsymbol{\omega}_0 = \boldsymbol{\omega}_0|_{s=s_0}\,, \qquad \boldsymbol{u}_0 = \boldsymbol{u}|_{s=s_0}\,. \tag{245}$$

$$\psi_0 = \begin{pmatrix} \hat{\boldsymbol{\omega}}_0 & \boldsymbol{u}_0 \\ 0 & 0 \end{pmatrix}. \tag{246}$$

$$\rho_0(t+\mathrm{d}t) = \rho_0(t)\mathrm{Oexp}(\psi_0(t+\cdot);\mathrm{d}t) \tag{247}$$

$$= \rho_0(t)\exp\left(\frac{\mathrm{d}t}{2}\left(\psi_0(t+\mathrm{d}t)+\psi_0(t)\right)\right) + O(\mathrm{d}t^3) \tag{248}$$

2.6.1 $s_0 = 0$

Evaluating (236),(237) at $s = 0$ results in

$$\boldsymbol{u}_0 = \frac{1}{c_\parallel}\left(\partial_s\lambda + f_a\right)\boldsymbol{e}_1 - \frac{B_\perp}{c_\perp}\partial_s^2\Omega_3\boldsymbol{e}_2 + \frac{B_\perp}{c_\perp}\partial_s^2\Omega_2\boldsymbol{e}_3 \tag{249}$$

$$\boldsymbol{\omega}_0 = \frac{1}{c_r}\left(B_\parallel\partial_s\Omega_1 + m_a\right)\boldsymbol{e}_1 - \frac{B_\perp}{c_\perp}\partial_s^3\Omega_2\boldsymbol{e}_2 - \frac{B_\perp}{c_\perp}\partial_s^3\Omega_3\boldsymbol{e}_3 \tag{250}$$

2.7 Boundary and initial conditions

To solve the equations of motion (232-234) and integrate the frame/cernterline we need initial data for

- Cross-section s_0 pose — $\rho(s_0, 0)$
- Curvature vector at each $s \in [0, L]$ — $\boldsymbol{\Omega}(s, 0)$

and boundary conditions

- $\times 2$ for Ω_1
- $\times 4$ for $\boldsymbol{\Omega}_\perp$
- $\times 2$ for λ

Since we're interested in the case where the ends of the body are unconstrained, the boundary conditions are given by (102)

$$\boldsymbol{f}|_{s=0,L} = 0 \tag{251}$$

$$B\boldsymbol{\Omega}|_{s=0,L} = 0 \tag{252}$$

Using (235) to eliminate $\boldsymbol{f}_\perp$, we find

$$\Omega_1 = 0, \boldsymbol{\Omega}_\perp = 0, \partial_s \boldsymbol{\Omega}_\perp = 0, \lambda = 0. \tag{253}$$

at $s = 0$ and $s = L$.

2.8 Nondimensional Equations 3D

In units with $L = 1, f_a^* = 1, c_\parallel = 1$ we have

$$\beta_\parallel = B_\parallel \tag{254}$$

$$\beta_\perp = B_\perp \tag{255}$$

$$\eta_\perp = c_\perp \tag{256}$$

$$\eta_r = c_r \tag{257}$$

$$U = 1 \tag{258}$$

$$\begin{aligned}
\partial_t \Omega_1 + \left[\frac{\beta_\perp - \beta_\parallel}{\eta_\perp}|\boldsymbol{\Omega}_\perp|^2 - \frac{\beta_\parallel}{\eta_r}\right]\partial_s^2 \Omega_1 &= G_1\left(\Omega_1, \partial_s\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \partial_s m_a\right) \\
\partial_t \boldsymbol{\Omega}_\perp + \frac{\beta_\perp}{\eta_\perp}\partial_s^4\boldsymbol{\Omega}_\perp + \frac{\beta_\perp - \beta_\parallel}{\eta_\perp}\boldsymbol{e}_1 \times \boldsymbol{\Omega}_\perp \partial_s^3 \Omega_1 &= G_\perp\left(\Omega_1, \ldots, \partial_s^2\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \ldots, \partial_s^2\lambda, f_a, \partial_s \right. \\
\partial_s^2 \lambda - \frac{1}{\eta_\perp}|\boldsymbol{\Omega}_\perp|^2\lambda &= G_\lambda\left(\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^2\boldsymbol{\Omega}_\perp, \partial_s f_a\right) \\
\partial_t f_a &= \left[1 - f_a^2\right] u_1 + \frac{\sigma}{U}\partial_s^2 f_a
\end{aligned} \tag{259}$$

$$\begin{aligned}
\Omega_1|_{s=0,L} &= 0 \\
\boldsymbol{\Omega}_\perp|_{s=0,L} &= 0 \\
\partial_s\boldsymbol{\Omega}_\perp|_{s=0,L} &= 0 \\
\lambda|_{s=0,L} &= 0 \\
\partial_s f_a|_{s=0,L} &= 0
\end{aligned} \tag{260}$$

3 2D Nondimensional equations (no twist/active moment)

- Curvature/Angular velocity – $\mathbf{\Omega} = \kappa \boldsymbol{e}_z$, $\boldsymbol{\omega} = \omega \boldsymbol{e}_z$
- Stiffness – $\beta = \frac{B_\perp}{f_a^* L^3}$
- Drag – $\eta = \frac{c_\perp}{c_\parallel} \approx 1.585$
- Active force constant, Length – $f_a^* = 1$, $L = 1$

Curvature evolution

$$\partial_t \kappa = -\frac{\beta}{\eta}\partial_s^4 \kappa + \frac{\beta}{3}\partial_s^2\left(\kappa^3\right) + \frac{1}{\eta}\partial_s^2\left(\kappa\lambda\right) + \partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) \tag{261}$$

$$\kappa|_{s=0,1} = 0, \quad \partial_s\kappa|_{s=0,1} = 0 \tag{262}$$

Tension equation

$$\partial_s^2\lambda - \frac{\kappa^2}{\eta}\lambda = -\partial_s f - \frac{\beta}{2}\partial_s^2\left(\kappa^2\right) - \frac{\beta}{\eta}\kappa\partial_s^2\kappa \tag{263}$$

$$\lambda|_{s=0,1} = 0 \tag{264}$$

Active force evolution

$$\partial_t f = \sigma\partial_s^2 f + \left(1 - f^2\right)\left(\beta\kappa\partial_s\kappa + \partial_s\lambda + f\right) \tag{265}$$

$$\partial_s f|_{s=0,1} = 0 \tag{266}$$

Tangential Velocity

$$u_\parallel = \frac{\beta}{2}\partial_s\left(\kappa^2\right) + \partial_s\lambda + f \tag{267}$$

Normal Velocity

$$u_\perp = -\frac{\beta\partial_s^2\kappa}{\eta} + \frac{\kappa\lambda}{\eta} \tag{268}$$

Angular Velocity

$$\omega = \beta\kappa^2\partial_s\kappa + \kappa\partial_s\lambda + \kappa f + \frac{\kappa\partial_s\lambda}{\eta} + \frac{\lambda\partial_s\kappa}{\eta} - \frac{\beta\partial_s^3\kappa}{\eta} \tag{269}$$

Some conserved stuff

$$\int_0^1 u_\parallel - f \,\mathrm{d}s = 0 \tag{270}$$

$$\int_0^1 u_\perp - \frac{1}{\eta}\kappa\lambda \,\mathrm{d}s = 0 \tag{271}$$

3.1 Steady state

Note: if $f = -1\Theta(s - 1/2) + 1$ you need to match bcs at $s = 1/2$

$$-\frac{\beta}{\eta}\partial_s^4\kappa + \frac{\beta}{3}\partial_s^2\left(\kappa^3\right) + \frac{1}{\eta}\partial_s^2\left(\kappa\lambda\right) + \partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) = 0 \tag{272}$$

$$\partial_s^4\kappa - \frac{\eta}{3}\partial_s^2\left(\kappa^3\right) - \frac{1}{\beta}\partial_s^2\left(\kappa\lambda\right) - \frac{\eta}{\beta}\partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) = 0 \tag{273}$$

$$\partial_s^3\kappa - \frac{\eta}{3}\partial_s\left(\kappa^3\right) - \frac{1}{\beta}\partial_s\left(\lambda\kappa\right) - \frac{\eta}{\beta}\left[\partial_s\lambda + f\right]\kappa = a \tag{274}$$

For some constante a. Evaluate at $s = 0, 1$ and apply boundary conditions to get

$$\partial_s^3\kappa\big|_{s=0,1} = a \tag{275}$$

3.2 Tension integral equation

The tension differential equation (263) is equivalent to the Fredholm equation

$$\lambda(s) = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z)f'(z)\,\mathrm{d}z - \frac{\beta}{2\eta}\kappa^2(s) + \frac{\beta}{\eta}\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z + \frac{1}{\eta}\int_0^1 G(s,z)\kappa^2(z) \tag{276}$$

where G is the Dirichlet Green function for the Laplace equation,

$$G(s,s') = \begin{cases} (s'-1)s, & \text{for } s < s' \\ s'(s-1), & \text{for } s > s' \end{cases} \tag{277}$$

$$= ss' - s\Theta(s'-s) - s'\Theta(s-s'). \tag{278}$$

To simplify notation, define the following:

$$\varepsilon = \frac{1}{\eta} \tag{279}$$

$$g_0(s) = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z)f'(z)\,\mathrm{d}z \tag{280}$$

$$g_1(s) = -\frac{\beta}{2}\kappa^2(s) + \beta\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z \tag{281}$$

$$K(x,y) = G(x,y)\kappa^2(y). \tag{282}$$

Equation (276) is then

$$|\lambda\rangle = |g_0\rangle + \varepsilon\,|g_1\rangle + \varepsilon\hat{K}\,|\lambda\rangle \tag{283}$$

has solution

$$|\lambda\rangle = |g_0\rangle + \sum_{j=1} \varepsilon^j \hat{K}^{j-1}\left[\hat{K}\,|g_0\rangle + |g_1\rangle\right]. \tag{284}$$

3.2.1 First few terms

$$\lambda_0 = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z)f'(z)\,\mathrm{d}z \tag{285}$$

$$\lambda_1 = -\frac{\beta}{2}\kappa^2(s) + \beta\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z - \frac{\beta}{2}\int_0^1 G(s,z)\kappa^4(z)\,\mathrm{d}z - \int_0^1\int_0^1 G(s,y)\kappa^2(y)G(y,z)f'(z)\,\mathrm{d}z\,\mathrm{d}y \tag{286}$$

3.2.2 $f(s) = 1 - 2\Theta(s - 1/2)$

If $f = -2\Theta(s-1/2)+1$ then $\partial_s = -2\delta(s-1/2)$ and

$$-\int_0^1 \mathrm{d}z\, G(s,z)\partial_z f(z) = 2G(s,1/2) \tag{287}$$

$$= s - 2s\Theta(1/2-s) - \Theta(s-1/2) \tag{288}$$

$$g_0(s) = -\frac{\beta}{2}\kappa^2(s) + s - 2s\Theta(1/2-s) - \Theta(s-1/2) \tag{289}$$

3.3 Constant curvature

$$f = -2\Theta(s-1/2)+1 \tag{290}$$

$$\partial_s f = -2\delta(s-1/2) \tag{291}$$

$$\partial_s^2\lambda - \frac{\kappa^2}{\eta}\lambda = 2\delta(s-1/2) \tag{292}$$

$$\partial_s\lambda(1/2^+) - \partial_s\lambda(1/2^-) = 2 \tag{293}$$

$$\lambda(0) = \lambda(1) = 0 \tag{294}$$

$$\lambda = \begin{cases} \frac{-\sqrt{\eta}}{\kappa\cosh\left(\frac{\kappa}{2\sqrt{\eta}}\right)}\sinh\left(\frac{\kappa}{\sqrt{\eta}}s\right), & \text{for } s < 1/2 \\ \frac{\sqrt{\eta}}{\kappa\cosh\left(\frac{\kappa}{2\sqrt{\eta}}\right)}\sinh\left(\frac{\kappa}{\sqrt{\eta}}(s-1)\right), & \text{for } s > 1/2 \end{cases} \tag{295}$$

$$= \begin{cases} -s + O(\kappa^2), & \text{for } s < 1/2 \\ s - 1 + O(\kappa^2), & \text{for } s > 1/2 \end{cases} \tag{296}$$

No swimming without $O(\kappa^2)$ terms when κ =constant?

$$\partial_s\lambda = 2\Theta(s-1/2) - 1 + O(\kappa^2) \tag{297}$$

$$u_\parallel = \partial_s\lambda + f = O(\kappa^2) \tag{298}$$

$$u_\perp = \frac{\kappa\lambda}{\eta} = \begin{cases} -\frac{\kappa s}{\eta} + O(\kappa^3), & \text{for } s < 1/2 \\ \frac{\kappa(s-1)}{\eta} + O(\kappa^3), & \text{for } s > 1/2 \end{cases} \tag{299}$$

3.4 Linear stability (2D)

Throw out all $O(\kappa^2)$ terms and let $\overline{f} = \int_0^1 f \, \mathrm{d}s$ be the average active force. Then, the tension satisfies,

$$\lambda = \overline{f} s - \int_0^s f \, \mathrm{d}s = -\int_0^s (f - \overline{f}) \, \mathrm{d}s \tag{300}$$

$$\partial_s \lambda + f = \overline{f}, \tag{301}$$

and the curvature is governed by

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa - \frac{1}{\eta} \partial_s^2 \left(\kappa \int_0^s f - \overline{f} \, \mathrm{d}s \right) + \overline{f} \partial_s \kappa, \tag{302}$$

Proof. Integrate $\partial_s^2 \lambda = -\partial_s f$ and use $\lambda(0) = \lambda(1) = 0$ to get $\lambda = \overline{f} s - \int_0^s f \, \mathrm{d}s$ and $\partial_s \lambda + f = \overline{f}$. Plug the tension expression into the linearized curvature equation

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa + \frac{1}{\eta} \partial_s^2 (\kappa \lambda) + \partial_s (\kappa [\partial_s \lambda + f]) \tag{303}$$

$$= -\frac{\beta}{\eta} \partial_s^4 \kappa + \frac{1}{\eta} \partial_s^2 \left(\left[\overline{f} s - \int_0^s f \, \mathrm{d}s \right] \kappa \right) + \overline{f} \partial_s \kappa \tag{304}$$

□

3.4.1 $\overline{f} = 0$

When the average active force vanishes, $\overline{f} = 0$,

$$\lambda = -\int_0^s f \, \mathrm{d}s \tag{305}$$

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa - \frac{1}{\eta} \partial_s^2 \left(\kappa \int_0^s f \, \mathrm{d}s \right) \tag{306}$$

$$\mathcal{L}[\kappa_n] = \chi_n \kappa_n = -\frac{\beta}{\eta} \partial_s^4 \kappa_n - \frac{1}{\eta} \partial_s^2 \left(\kappa_n \int_0^s f \, \mathrm{d}s \right) \tag{307}$$

3.4.2 Piecewise constant active force

$$f = \begin{cases} 1 & , \text{ when } s < 1/2 \\ -1 & , \text{ when } s > 1/2 \end{cases} \tag{308}$$

$$\lambda = \begin{cases} -s & , \text{ when } s < 1/2 \\ s-1 & , \text{ when } s > 1/2 \end{cases} \tag{309}$$

$$\partial_t \kappa = \begin{cases} -\frac{\beta}{\eta}\partial_s^2 \left(\partial_s^2 + \frac{s}{\beta}\right)\kappa & , \text{ when } s < 1/2 \\ -\frac{\beta}{\eta}\partial_s^2 \left(\partial_s^2 + \frac{1-s}{\beta}\right)\kappa & , \text{ when } s > 1/2 \end{cases} \tag{310}$$

3.4.3 Regularized piecewise constant active force

$$f = -\tanh(a(s-1/2)) \tag{311}$$

$$f \to -2\Theta(s-1/2) + 1 \text{ as } a \to \infty \tag{312}$$

Tension

$$\partial_s^2 \lambda = -\partial_s f = a\,\mathrm{sech}^2(a(s-1/2)) \tag{313}$$

$$\partial_s \lambda = \tanh(a(s-1/2)) + b \tag{314}$$

$$\lambda = \ln(\cosh(a(s-1/2)))/a + bs + c \tag{315}$$

$$\lambda|_{s=0,1} = 0 \Rightarrow b = 0, c = -\ln(\cosh(a/2))/a \tag{316}$$

$$\lambda = \frac{1}{a}\ln\left(\frac{\cosh(a(s-1/2))}{\cosh(a/2)}\right) \tag{317}$$

since $\partial_s \lambda + f = 0$, the curvature equation reduces to

$$\partial_t \kappa = -\frac{\beta}{\eta}\partial_s^4 \kappa + \frac{1}{\eta}\partial_s^2 \left(\frac{1}{a}\ln\left(\frac{\cosh(a(s-1/2))}{\cosh(a/2)}\right)\kappa\right) \tag{318}$$

3.5 Linear reduced order model

-

4 3D Sympy expressions

Twist

$$
\begin{aligned}
&-\frac{M\partial_s f}{\eta_r}+\frac{\Omega_0^3\Omega_1^2\beta_\parallel}{\eta}-\frac{\Omega_0^3\Omega_1^2\beta_\perp}{\eta}+\frac{\Omega_0^3\Omega_2^2\beta_\parallel}{\eta}-\frac{\Omega_0^3\Omega_2^2\beta_\perp}{\eta}\\
&+\frac{2\Omega_0^2\Omega_1\beta_\parallel\partial_s\Omega_2}{\eta}-\frac{3\Omega_0^2\Omega_1\beta_\perp\partial_s\Omega_2}{\eta}-\frac{2\Omega_0^2\Omega_2\beta_\parallel\partial_s\Omega_1}{\eta}+\frac{3\Omega_0^2\Omega_2\beta_\perp\partial_s\Omega_1}{\eta}-\frac{\Omega_0\Omega_1^2\lambda}{\eta}\\
&-\frac{\Omega_0\Omega_1\beta_\parallel\partial_s^2\Omega_1}{\eta}+\frac{3\Omega_0\Omega_1\beta_\perp\partial_s^2\Omega_1}{\eta}-\frac{\Omega_0\Omega_2^2\lambda}{\eta}-\frac{\Omega_0\Omega_2\beta_\parallel\partial_s^2\Omega_2}{\eta}+\frac{3\Omega_0\Omega_2\beta_\perp\partial_s^2\Omega_2}{\eta}\\
&-\frac{\Omega_1^2\beta_\parallel\partial_s^2\Omega_0}{\eta}+\frac{\Omega_1^2\beta_\perp\partial_s^2\Omega_0}{\eta}-\frac{2\Omega_1\beta_\parallel\partial_s\Omega_0\partial_s\Omega_1}{\eta}+\frac{3\Omega_1\beta_\perp\partial_s\Omega_0\partial_s\Omega_1}{\eta}+\frac{\Omega_1\beta_\perp\partial_s^3\Omega_2}{\eta}\\
&-\frac{\Omega_1\lambda\partial_s\Omega_2}{\eta}-\frac{\Omega_2^2\beta_\parallel\partial_s^2\Omega_0}{\eta}+\frac{\Omega_2^2\beta_\perp\partial_s^2\Omega_0}{\eta}-\frac{2\Omega_2\beta_\parallel\partial_s\Omega_0\partial_s\Omega_2}{\eta}+\frac{3\Omega_2\beta_\perp\partial_s\Omega_0\partial_s\Omega_2}{\eta}\\
&-\frac{\Omega_2\beta_\perp\partial_s^3\Omega_1}{\eta}+\frac{\Omega_2\lambda\partial_s\Omega_1}{\eta}-\frac{\beta_\parallel\partial_s^2\Omega_0}{\eta_r}+\partial_t\Omega_0=0
\end{aligned}
\tag{319}
$$

Curvature1

$$
\begin{aligned}
&-\frac{M\Omega_2 f}{\eta_r}-\frac{\Omega_0^4\Omega_1\beta_\parallel}{\eta}+\frac{\Omega_0^4\Omega_1\beta_\perp}{\eta}-\frac{3\Omega_0^3\beta_\parallel\partial_s\Omega_2}{\eta}+\frac{4\Omega_0^3\beta_\perp\partial_s\Omega_2}{\eta}\\
&+\frac{\Omega_0^2\Omega_1\lambda}{\eta}-\frac{6\Omega_0^2\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta}+\frac{6\Omega_0^2\Omega_2\beta_\perp\partial_s\Omega_0}{\eta}+\frac{3\Omega_0^2\beta_\parallel\partial_s^2\Omega_1}{\eta}-\frac{6\Omega_0^2\beta_\perp\partial_s^2\Omega_1}{\eta}\\
&+\Omega_0\Omega_1\Omega_2\beta_\perp\partial_s\Omega_1+\frac{4\Omega_0\Omega_1\beta_\parallel\partial_s^2\Omega_0}{\eta}-\frac{4\Omega_0\Omega_1\beta_\perp\partial_s^2\Omega_0}{\eta}+\Omega_0\Omega_2^2\beta_\perp\partial_s\Omega_2+\Omega_0\Omega_2\partial_s\lambda\\
&+\Omega_0\Omega_2 f+\frac{2\Omega_0\Omega_2\partial_s\lambda}{\eta}+\frac{9\Omega_0\beta_\parallel\partial_s\Omega_0\partial_s\Omega_1}{\eta}+\frac{\Omega_0\beta_\parallel\partial_s^3\Omega_2}{\eta}-\frac{12\Omega_0\beta_\perp\partial_s\Omega_0\partial_s\Omega_1}{\eta}\\
&-\frac{4\Omega_0\beta_\perp\partial_s^3\Omega_2}{\eta}+\frac{2\Omega_0\lambda\partial_s\Omega_2}{\eta}-\Omega_1^2\beta_\perp\partial_s^2\Omega_1-\Omega_1\Omega_2\beta_\perp\partial_s^2\Omega_2+\frac{3\Omega_1\beta_\parallel\partial_s\Omega_0^2}{\eta}\\
&-2\Omega_1\beta_\perp\partial_s\Omega_1^2-\Omega_1\beta_\perp\partial_s\Omega_2^2-\frac{3\Omega_1\beta_\perp\partial_s\Omega_0^2}{\eta}-\Omega_1\partial_s^2\lambda-\Omega_1\partial_s f\\
&-\frac{\Omega_1\partial_s^2\lambda}{\eta}-\frac{\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta_r}+\frac{\Omega_2\beta_\parallel\partial_s^3\Omega_0}{\eta}-\Omega_2\beta_\perp\partial_s\Omega_1\partial_s\Omega_2-\frac{\Omega_2\beta_\perp\partial_s^3\Omega_0}{\eta}\\
&+\frac{\Omega_2\lambda\partial_s\Omega_0}{\eta}+\frac{3\beta_\parallel\partial_s\Omega_0\partial_s^2\Omega_2}{\eta}+\frac{3\beta_\parallel\partial_s\Omega_2\partial_s^2\Omega_0}{\eta}-\frac{6\beta_\perp\partial_s\Omega_0\partial_s^2\Omega_2}{\eta}-\frac{4\beta_\perp\partial_s\Omega_2\partial_s^2\Omega_0}{\eta}\\
&+\frac{\beta_\perp\partial_s^4\Omega_1}{\eta}-\partial_s\Omega_1\partial_s\lambda-\partial_s\Omega_1 f+\partial_t\Omega_1-\frac{\lambda\partial_s^2\Omega_1}{\eta}\\
&-\frac{2\partial_s\Omega_1\partial_s\lambda}{\eta}=0
\end{aligned}
\tag{320}
$$

Curvature2

$$\frac{M\Omega_1 f}{\eta_r} - \frac{\Omega_0^4\Omega_2\beta_\parallel}{\eta} + \frac{\Omega_0^4\Omega_2\beta_\perp}{\eta} + \frac{3\Omega_0^3\beta_\parallel\partial_s\Omega_1}{\eta} - \frac{4\Omega_0^3\beta_\perp\partial_s\Omega_1}{\eta}$$
$$+ \frac{6\Omega_0^2\Omega_1\beta_\parallel\partial_s\Omega_0}{\eta} - \frac{6\Omega_0^2\Omega_1\beta_\perp\partial_s\Omega_0}{\eta} + \frac{\Omega_0^2\Omega_2\lambda}{\eta} + \frac{3\Omega_0^2\beta_\parallel\partial_s^2\Omega_2}{\eta} - \frac{6\Omega_0^2\beta_\perp\partial_s^2\Omega_2}{\eta}$$
$$- \Omega_0\Omega_1^2\beta_\perp\partial_s\Omega_1 - \Omega_0\Omega_1\Omega_2\beta_\perp\partial_s\Omega_2 - \Omega_0\Omega_1\partial_s\lambda - \Omega_0\Omega_1 f - \frac{2\Omega_0\Omega_1\partial_s\lambda}{\eta}$$
$$+ \frac{4\Omega_0\Omega_2\beta_\parallel\partial_s^2\Omega_0}{\eta} - \frac{4\Omega_0\Omega_2\beta_\perp\partial_s^2\Omega_0}{\eta} + \frac{9\Omega_0\beta_\parallel\partial_s\Omega_0\partial_s\Omega_2}{\eta} - \frac{\Omega_0\beta_\parallel\partial_s^3\Omega_1}{\eta} - \frac{12\Omega_0\beta_\perp\partial_s\Omega_0\partial_s\Omega_2}{\eta}$$
$$+ \frac{4\Omega_0\beta_\perp\partial_s^3\Omega_1}{\eta} - \frac{2\Omega_0\lambda\partial_s\Omega_1}{\eta} - \Omega_1\Omega_2\beta_\perp\partial_s^2\Omega_1 + \frac{\Omega_1\beta_\parallel\partial_s\Omega_0}{\eta_r} - \frac{\Omega_1\beta_\parallel\partial_s^3\Omega_0}{\eta}$$
$$- \Omega_1\beta_\perp\partial_s\Omega_1\partial_s\Omega_2 + \frac{\Omega_1\beta_\perp\partial_s^3\Omega_0}{\eta} - \frac{\Omega_1\lambda\partial_s\Omega_0}{\eta} - \Omega_2^2\beta_\perp\partial_s^2\Omega_2 + \frac{3\Omega_2\beta_\parallel\partial_s\Omega_0^2}{\eta}$$
$$- \Omega_2\beta_\perp\partial_s\Omega_1^2 - 2\Omega_2\beta_\perp\partial_s\Omega_2^2 - \frac{3\Omega_2\beta_\perp\partial_s\Omega_0^2}{\eta} - \Omega_2\partial_s^2\lambda - \Omega_2\partial_s f$$
$$- \frac{\Omega_2\partial_s^2\lambda}{\eta} - \frac{3\beta_\parallel\partial_s\Omega_0\partial_s^2\Omega_1}{\eta} - \frac{3\beta_\parallel\partial_s\Omega_1\partial_s^2\Omega_0}{\eta} + \frac{6\beta_\perp\partial_s\Omega_0\partial_s^2\Omega_1}{\eta} + \frac{4\beta_\perp\partial_s\Omega_1\partial_s^2\Omega_0}{\eta}$$
$$+ \frac{\beta_\perp\partial_s^4\Omega_2}{\eta} - \partial_s\Omega_2\partial_s\lambda - \partial_s\Omega_2 f + \partial_t\Omega_2 - \frac{\lambda\partial_s^2\Omega_2}{\eta}$$
$$- \frac{2\partial_s\Omega_2\partial_s\lambda}{\eta} = 0 \tag{321}$$

Active force

$$- \partial_s^2 f\sigma + \partial_t f - \left(1 - f^2\right)\left(\Omega_1\beta_\perp\partial_s\Omega_1 + \Omega_2\beta_\perp\partial_s\Omega_2 + \partial_s\lambda + f\right) = 0 \tag{322}$$

Tension

$$\frac{\Omega_0^2\Omega_1^2\beta_\parallel}{\eta} - \frac{\Omega_0^2\Omega_1^2\beta_\perp}{\eta} + \frac{\Omega_0^2\Omega_2^2\beta_\parallel}{\eta} - \frac{\Omega_0^2\Omega_2^2\beta_\perp}{\eta} + \frac{\Omega_0\Omega_1\beta_\parallel\partial_s\Omega_2}{\eta}$$
$$- \frac{2\Omega_0\Omega_1\beta_\perp\partial_s\Omega_2}{\eta} - \frac{\Omega_0\Omega_2\beta_\parallel\partial_s\Omega_1}{\eta} + \frac{2\Omega_0\Omega_2\beta_\perp\partial_s\Omega_1}{\eta} - \frac{\Omega_1^2\lambda}{\eta} + \Omega_1\beta_\perp\partial_s^2\Omega_1$$
$$+ \frac{\Omega_1\beta_\perp\partial_s^2\Omega_1}{\eta} - \frac{\Omega_2^2\lambda}{\eta} + \Omega_2\beta_\perp\partial_s^2\Omega_2 + \frac{\Omega_2\beta_\perp\partial_s^2\Omega_2}{\eta} + \beta_\perp\partial_s\Omega_1^2$$
$$+ \beta_\perp\partial_s\Omega_2^2 + \partial_s^2\lambda + \partial_s f = 0 \tag{323}$$

Angular velocity0

$$\omega_0 = \frac{Mf}{\eta_r} + \frac{\beta_\parallel \partial_s \Omega_0}{\eta_r} \tag{324}$$

Angular velocity1

$$\begin{aligned}\omega_1 =& \frac{\Omega_0^3 \Omega_2 \beta_\parallel}{\eta} - \frac{\Omega_0^3 \Omega_2 \beta_\perp}{\eta} - \frac{2\Omega_0^2 \beta_\parallel \partial_s \Omega_1}{\eta} + \frac{3\Omega_0^2 \beta_\perp \partial_s \Omega_1}{\eta} - \frac{3\Omega_0 \Omega_1 \beta_\parallel \partial_s \Omega_0}{\eta} \\ &+ \frac{3\Omega_0 \Omega_1 \beta_\perp \partial_s \Omega_0}{\eta} - \frac{\Omega_0 \Omega_2 \lambda}{\eta} - \frac{\Omega_0 \beta_\parallel \partial_s^2 \Omega_2}{\eta} + \frac{3\Omega_0 \beta_\perp \partial_s^2 \Omega_2}{\eta} + \Omega_1^2 \beta_\perp \partial_s \Omega_1 \\ &+ \Omega_1 \Omega_2 \beta_\perp \partial_s \Omega_2 + \Omega_1 \partial_s \lambda + \Omega_1 f + \frac{\Omega_1 \partial_s \lambda}{\eta} - \frac{\Omega_2 \beta_\parallel \partial_s^2 \Omega_0}{\eta} \\ &+ \frac{\Omega_2 \beta_\perp \partial_s^2 \Omega_0}{\eta} - \frac{2\beta_\parallel \partial_s \Omega_0 \partial_s \Omega_2}{\eta} + \frac{3\beta_\perp \partial_s \Omega_0 \partial_s \Omega_2}{\eta} - \frac{\beta_\perp \partial_s^3 \Omega_1}{\eta} + \frac{\lambda \partial_s \Omega_1}{\eta}\end{aligned} \tag{325}$$

Angular velocity2

$$\begin{aligned}\omega_2 =& - \frac{\Omega_0^3 \Omega_1 \beta_\parallel}{\eta} + \frac{\Omega_0^3 \Omega_1 \beta_\perp}{\eta} - \frac{2\Omega_0^2 \beta_\parallel \partial_s \Omega_2}{\eta} + \frac{3\Omega_0^2 \beta_\perp \partial_s \Omega_2}{\eta} + \frac{\Omega_0 \Omega_1 \lambda}{\eta} \\ &- \frac{3\Omega_0 \Omega_2 \beta_\parallel \partial_s \Omega_0}{\eta} + \frac{3\Omega_0 \Omega_2 \beta_\perp \partial_s \Omega_0}{\eta} + \frac{\Omega_0 \beta_\parallel \partial_s^2 \Omega_1}{\eta} - \frac{3\Omega_0 \beta_\perp \partial_s^2 \Omega_1}{\eta} + \Omega_1 \Omega_2 \beta_\perp \partial_s \Omega_1 \\ &+ \frac{\Omega_1 \beta_\parallel \partial_s^2 \Omega_0}{\eta} - \frac{\Omega_1 \beta_\perp \partial_s^2 \Omega_0}{\eta} + \Omega_2^2 \beta_\perp \partial_s \Omega_2 + \Omega_2 \partial_s \lambda + \Omega_2 f \\ &+ \frac{\Omega_2 \partial_s \lambda}{\eta} + \frac{2\beta_\parallel \partial_s \Omega_0 \partial_s \Omega_1}{\eta} - \frac{3\beta_\perp \partial_s \Omega_0 \partial_s \Omega_1}{\eta} - \frac{\beta_\perp \partial_s^3 \Omega_2}{\eta} + \frac{\lambda \partial_s \Omega_2}{\eta}\end{aligned} \tag{326}$$

Linear velocity0

$$u_0 = \Omega_1 \beta_\perp \partial_s \Omega_1 + \Omega_2 \beta_\perp \partial_s \Omega_2 + \partial_s \lambda + f \tag{327}$$

Linear velocity1

$$\begin{aligned}u_1 =& - \frac{\Omega_0^2 \Omega_2 \beta_\parallel}{\eta} + \frac{\Omega_0^2 \Omega_2 \beta_\perp}{\eta} + \frac{\Omega_0 \beta_\parallel \partial_s \Omega_1}{\eta} - \frac{2\Omega_0 \beta_\perp \partial_s \Omega_1}{\eta} + \frac{\Omega_1 \beta_\parallel \partial_s \Omega_0}{\eta} \\ &- \frac{\Omega_1 \beta_\perp \partial_s \Omega_0}{\eta} + \frac{\Omega_2 \lambda}{\eta} - \frac{\beta_\perp \partial_s^2 \Omega_2}{\eta}\end{aligned} \tag{328}$$

Linear velocity2

$$
\begin{aligned}
u_2 =& \frac{\Omega_0^2\Omega_1\beta_\parallel}{\eta} - \frac{\Omega_0^2\Omega_1\beta_\perp}{\eta} + \frac{\Omega_0\beta_\parallel\partial_s\Omega_2}{\eta} - \frac{2\Omega_0\beta_\perp\partial_s\Omega_2}{\eta} - \frac{\Omega_1\lambda}{\eta} \\
&+ \frac{\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta} - \frac{\Omega_2\beta_\perp\partial_s\Omega_0}{\eta} + \frac{\beta_\perp\partial_s^2\Omega_1}{\eta}
\end{aligned} \tag{329}
$$

4.0.1 3D Linearized Equations

$$
\partial_t\Omega_0 - \frac{\beta_\parallel}{\eta_r}\partial_s^2\Omega_0 - \frac{M}{\eta_r}\partial_s f = 0 \tag{330}
$$

$$
\partial_t\Omega_1 + \frac{\beta_\perp}{\eta_\perp}\partial_s^4\Omega_1 = 0 \tag{331}
$$

$$
\partial_t\Omega_2 + \frac{\beta_\perp}{\eta_\perp}\partial_s^4\Omega_2 = 0 \tag{332}
$$

$$
\partial_t f - \sigma\partial_s^2 f - f - \partial_s\lambda = 0 \tag{333}
$$

$$
\partial_s^2\lambda + \partial_s f = 0 \tag{334}
$$

5 Trajectories

...

6 3D Numerics

The equations of motion (259) can be written in the form

$$
\begin{aligned}
\partial_t\Omega_0 - \dot{\Omega}_0\left(\Omega_0, \partial_s\Omega_0, \partial_s^2\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \partial_s f, \lambda\right) &= 0 \\
\partial_t\Omega_1 - \dot{\Omega}_1\left(\Omega_0, \ldots, \partial_s^3\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^4\boldsymbol{\Omega}_\perp, f, \partial_s f, \lambda, \ldots, \partial_s^2\lambda\right) &= 0 \\
\partial_t\Omega_2 - \dot{\Omega}_2\left(\Omega_0, \ldots, \partial_s^3\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^4\boldsymbol{\Omega}_\perp, f, \partial_s f, \lambda, \ldots, \partial_s^2\lambda\right) &= 0 \\
\partial_t f - \dot{f}(\boldsymbol{\Omega}_\perp, \partial_s\boldsymbol{\Omega}_\perp, f, \partial_s^2 f, \partial_s\lambda) &= 0 \\
\Lambda\left(\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^2\boldsymbol{\Omega}_\perp, \partial_s f, \lambda, \partial_s^2\lambda\right) &= 0
\end{aligned} \tag{335}
$$

$$
\begin{aligned}
\Omega_0|_{s=0,L}\,,\ \Omega_1|_{s=0,L}\,,\ \partial_s\Omega_1|_{s=0,L}\,,\ \Omega_2|_{s=0,L} &= 0 \\
\partial_s\Omega_2|_{s=0,L}\,,\ \partial_s f|_{s=0,L}\,,\ \lambda|_{s=0,L} &= 0
\end{aligned} \tag{336}
$$

To simplify some calculations, we're just going to think of $\dot{\Omega}_0, \dot{\Omega}_1, \dot{\Omega}_2, \dot{f}, \Lambda$ as functions of all variables and their first four derivatives. That is, we treat them as maps

$$\begin{aligned} \dot{\Omega}_0 &: \mathbb{R}^{25} \to \mathbb{R} \\ \dot{\Omega}_1 &: \mathbb{R}^{25} \to \mathbb{R} \\ \dot{\Omega}_2 &: \mathbb{R}^{25} \to \mathbb{R} \\ \dot{f} &: \mathbb{R}^{25} \to \mathbb{R} \\ \Lambda &: \mathbb{R}^{25} \to \mathbb{R} \end{aligned} \tag{337}$$

More notation

$$X^p := \begin{cases} \Omega_0, & p = 0 \\ \Omega_1, & p = 1 \\ \Omega_2, & p = 2 \\ f, & p = 3 \\ \lambda, & p = 4 \end{cases} \tag{338}$$

Denote canonical basis vectors for $\mathbb{R}^5$ by $|p\rangle$ and the vector samples at s by

$$X := \sum_{p \in \mathbb{Z}_5} X^p \, |p\rangle \, . \tag{339}$$

For each $p \in \mathbb{Z}^5$, define

$$G^p : \mathbb{R}^{25} \to \mathbb{R} \tag{340}$$

$$\tag{341}$$

$$\begin{aligned} G^0(y_q^p) &= -\dot{\Omega}_0(y_q^p) \\ G^1(y_q^p) &= -\dot{\Omega}_1(y_q^p) \\ G^2(y_q^p) &= -\dot{\Omega}_2(y_q^p) \\ G^3(y_q^p) &= -\dot{f}(y_q^p) \\ G^4(y_q^p) &= \Lambda(y_q^p) \end{aligned} \tag{342}$$

where y_q^p is shorthand for

$$(y_0^0, y_0^1, \ldots, y_0^4, \ldots y_4^4, \ldots y_4^4) \in \mathbb{R}^{25} \tag{343}$$

The system (335) can then be written

$$(1 - \delta_4^p)\partial_t X^p + G^p(X, \ldots \partial_s^4 X) = 0, \text{ for } p \in \mathbb{Z}_5 \tag{344}$$

6.1 BDF2 with Newton

This section has a bunch of calculations you need to solve the discretization of (344) using Newton's method.

- Arclength index – $s \in \{-2, -1, 0, \ldots, N-1, N, N+1\}$
- Python index$j \in \mathbb{Z}_{N+4}$, $j \equiv s(\bmod N+4)$
- Variable index – $p \in \mathbb{Z}_5$
- Derivative index – $q \in \mathbb{Z}_6$

Let $\mathcal{H}$ be a $(5N+20)$-dimensional real Hilbert space with orthonormal basis,

$$|sp\rangle = |s\rangle \otimes |p\rangle \text{ for } -2 \leq s \leq N+1 \text{ and } p \in \mathbb{Z}_5. \tag{345}$$

Write state vectors as

$$|X\rangle = \sum_{s=-2}^{N+1} \sum_{p \in \mathbb{Z}_5} X^{sp} |sp\rangle \tag{346}$$

where X^{sp} represents the sample of the p-th variable at grid point s. When neccessary, we'll also add an additional time superscript,

$$X^{spt}, X^{sp,t-1}, \text{etc} \ldots \tag{347}$$

Define the 2nd order accurate difference operators

$$D^{(q)} = \sum_{|s-s'| \leq 2} \sum_{p \in \mathbb{Z}_5} D^{(q)}_{ss'} |sp\rangle \langle s'p| \tag{348}$$

$$D^{(q)}_{ss'} \delta_{pp'} = \langle sp| D^{(q)} |s'p'\rangle \tag{349}$$

$$D^{(0)}_{ss'} = \delta_{ss'} \tag{350}$$

$$D^{(1)}_{ss'} = -\frac{1}{2\,\mathrm{d}s}\delta_{s-1,s'} + \frac{1}{2\,\mathrm{d}s}\delta_{s+1,s'} \tag{351}$$

$$D^{(2)}_{ss'} = \frac{1}{\mathrm{d}s^2}\delta_{s-1,s'} - \frac{2}{\mathrm{d}s^2}\delta_{s,s'} + \frac{1}{\mathrm{d}s^2}\delta_{s+1,s'} \tag{352}$$

$$D^{(3)}_{ss'} = -\frac{1}{2\,\mathrm{d}s^3}\delta_{s-2,s'} + \frac{1}{\mathrm{d}s^3}\delta_{s-1,s'} - \frac{1}{\mathrm{d}s^3}\delta_{s+1,s'} + \frac{1}{2\,\mathrm{d}s^3}\delta_{s+2,s'} \tag{353}$$

$$D^{(4)}_{ss'} = \frac{1}{\mathrm{d}s^4}\delta_{s-2,s'} - \frac{4}{\mathrm{d}s^4}\delta_{s-1,s'} + \frac{6}{\mathrm{d}s^4}\delta_{s,s'} - \frac{4}{\mathrm{d}s^4}\delta_{s+1,s'} + \frac{1}{\mathrm{d}s^4}\delta_{s+2,s'} \tag{354}$$

$$D_{tt'} = \frac{1}{2\,\mathrm{d}t}\delta_{t-2,t'} - \frac{2}{\mathrm{d}t}\delta_{t-1,t'} + \frac{3}{2\,\mathrm{d}t}\delta_{t,t'} \tag{355}$$

and write finite differences of variables as

$$X^{sp}_q := \begin{cases} \sum\limits_{|s'-s|\leq 2} D^{(q)}_{ss'} X^{s'p}, & \text{for } q \in \mathbb{Z}_5 \\ \frac{1}{2\mathrm{d}t} X^{s,t-2,p} - \frac{2}{\mathrm{d}t} X^{s,t-1,p} + \frac{3}{2\mathrm{d}t} X^{s,t,p}, & \text{for } q = 5. \end{cases} \tag{356}$$

For $q \in \mathbb{Z}_5$

$$X^{sp}_q = X^{sp}_q(X^{s-2,p}, X^{s-1,p}, X^{s,p}, X^{s+1,p}, X^{s+2,p}). \tag{357}$$

Write $G^p(...X^{sp'}_q...)$ as

$$G^{sp} := G^p(...X^{sp'}_q...). \tag{358}$$

$$G^{sp} = G^{sp}(X^{s-2,p}, X^{s-1,p}, X^{s,p}, X^{s+1,p}, X^{s+2,p}) \tag{359}$$

The discretization of (344) is

$$(1-\delta^p_4)X^{sp}_5 + G^{sp} = 0, \text{ for } p \in \mathbb{Z}_5 \tag{360}$$

The state at time t is a root of the function

$$\begin{aligned} F : \mathcal{H} \to & \mathcal{H} \\ |X\rangle \mapsto & |F(X)\rangle \\ & = \sum_{s=-2}^{N+1} F^{sp} |sp\rangle \\ & = \sum_{s=0}^{N-1} \sum_{p\in\mathbb{Z}_5} \left[(1-\delta^p_4)X^{sp}_5 + G^{sp}\right] |sp\rangle \\ & + |BC\rangle \end{aligned} \tag{361}$$

where $|BC\rangle$ is defined by

$$
\begin{aligned}
|BC\rangle &= \sum_{s=-2}^{-1} \sum_{p\in\mathbb{Z}_5} B^{sp} \, |sp\rangle + \sum_{s=N}^{N+1} \sum_{p\in\mathbb{Z}_5} B^{sp} \, |sp\rangle \\
BC^{sp} &= \langle sp|BC\rangle
\end{aligned}
\tag{362}
$$

$$
BC^{-2,p} = \begin{cases}
X^{-2,0} = \Omega_0(-2) & \text{for } p = 0 \\
X^{1,1} - X^{-1,1} = \mathrm{d}s \, \partial_s \Omega_1(0) & \text{for } p = 1 \\
X^{1,2} - X^{-1,2} = \mathrm{d}s \, \partial_s \Omega_2(0) & \text{for } p = 2 \\
X^{-2,3} = f(-2) & \text{for } p = 3 \\
X^{-2,4} = \lambda(-2) & \text{for } p = 4
\end{cases}
\tag{363}
$$

$$
BC^{-1,p} = \begin{cases}
X^{0,0} = \Omega_0(0) & \text{for } p = 0 \\
X^{0,1} = \Omega_1(0) & \text{for } p = 1 \\
X^{0,2} = \Omega_2(0) & \text{for } p = 2 \\
X^{1,3} - X^{-1,3} = \mathrm{d}s \, \partial_s f(0) & \text{for } p = 3 \\
X^{0,4} = \lambda(0) & \text{for } p = 4
\end{cases}
\tag{364}
$$

$$
BC^{N,p} = \begin{cases}
X^{N-1,0} = \Omega_0(N-1) & \text{for } p = 0 \\
X^{N-1,1} = \Omega_1(N-1) & \text{for } p = 1 \\
X^{N-1,2} = \Omega_2(N-1) & \text{for } p = 2 \\
X^{N,3} - X^{N-2,3} = \mathrm{d}s \, \partial_s f(N-1) & \text{for } p = 3 \\
X^{N-1,4} = \lambda(N-1) & \text{for } p = 4
\end{cases}
\tag{365}
$$

$$
BC^{N+1,p} = \begin{cases}
X^{N+1,0} = \Omega_0(N+1) & \text{for } p = 0 \\
X^{N,1} - X^{N-2,1} = \mathrm{d}s \, \partial_s \Omega_1(N-1) & \text{for } p = 1 \\
X^{N,2} - X^{N-2,2} = \mathrm{d}s \, \partial_s \Omega_2(N-1) & \text{for } p = 2 \\
X^{N+1,3} = f(N+1) & \text{for } p = 3 \\
X^{N+1,4} = \lambda(N+1) & \text{for } p = 4
\end{cases}
\tag{366}
$$

Some things you need to calculate Jacobian:

$$\frac{\partial X_5^{sp}}{\partial X^{s'p'}} = \frac{3}{2\,\mathrm{dt}}\delta^s_{s'}\delta^p_{p'} \tag{367}$$

$$\frac{\partial X_q^{sp}}{\partial X^{s'p'}} = \frac{\partial}{\partial X^{s'p'}}\sum_{s''} D^{(q)}_{ss''}X^{s''p} \tag{368}$$

$$= D^{(q)}_{ss'}\delta^p_{p'} \tag{369}$$

$$\frac{\partial G^{sp}}{\partial X^{s'p'}} = \sum_{p'',q}\frac{\partial G^p}{\partial X_q^{sp''}}\frac{\partial X_q^{sp''}}{\partial X^{s'p'}} \tag{370}$$

$$= \sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{ss'} \tag{371}$$

Note that $D^{(q)}_{ss'} = 0$ and $\frac{\partial G^{sp}}{\partial X^{s'p'}} = 0$ when $|s - s'| > 2$.

$$\frac{\partial}{\partial X^{s'p'}}BC^{-2,p} = \begin{cases} \delta^{-2}_{s'}\delta^0_{p'} & \text{for } p = 0 \\ \delta^1_{s'}\delta^1_{p'} - \delta^{-1}_{s'}\delta^1_{p'} & \text{for } p = 1 \\ \delta^1_{s'}\delta^2_{p'} - \delta^{-1}_{s'}\delta^2_{p'} & \text{for } p = 2 \\ \delta^{-2}_{s'}\delta^3_{p'} & \text{for } p = 3 \\ \delta^{-2}_{s'}\delta^4_{p'} & \text{for } p = 4 \end{cases} \tag{372}$$

$$\frac{\partial}{\partial X^{s'p'}}BC^{-1,p} = \begin{cases} \delta^0_{s'}\delta^0_{p'} & \text{for } p = 0 \\ \delta^0_{s'}\delta^1_{p'} & \text{for } p = 1 \\ \delta^0_{s'}\delta^2_{p'} & \text{for } p = 2 \\ \delta^1_{s'}\delta^3_{p'} - \delta^{-1}_{s'}\delta^3_{p'} & \text{for } p = 3 \\ \delta^0_{s'}\delta^4_{p'} & \text{for } p = 4 \end{cases} \tag{373}$$

$$\frac{\partial}{\partial X^{s'p'}}BC^{Np} = \begin{cases} \delta^{N-1}_{s'}\delta^0_{p'} & \text{for } p = 0 \\ \delta^{N-1}_{s'}\delta^1_{p'} & \text{for } p = 1 \\ \delta^{N-1}_{s'}\delta^2_{p'} & \text{for } p = 2 \\ \delta^{N}_{s'}\delta^3_{p'} - \delta^{N-2}_{s'}\delta^3_{p'} & \text{for } p = 3 \\ \delta^{N-1}_{s'}\delta^4_{p'} & \text{for } p = 4 \end{cases} \tag{374}$$

$$\frac{\partial}{\partial X^{s'p'}} BC^{N+1,p} = \begin{cases} \delta^{N+1}_{s'} \delta^0_{p'} & \text{for } p = 0 \\ \delta^N_{s'} \delta^1_{p'} - \delta^{N-2}_{s'} \delta^1_{p'} & \text{for } p = 1 \\ \delta^N_{s'} \delta^2_{p'} - \delta^{N-2}_{s'} \delta^2_{p'} & \text{for } p = 2 \\ \delta^{N+1}_{s'} \delta^3_{p'} & \text{for } p = 3 \\ \delta^{N+1}_{s'} \delta^4_{p'} & \text{for } p = 4 \end{cases} \tag{375}$$

$$\frac{\partial BC^{sp}}{\partial X^{s'p'}} = \delta^p_{p'} \frac{\partial BC^{sp}}{\partial X^{s'p}} \tag{376}$$

$$\frac{\partial}{\partial X^{s'p'}} \left[(1 - \delta^p_4) X^{sp}_5 + G^{sp} \right] = (1 - \delta^p_4) \frac{3}{2\,\mathrm{d}t} \delta^s_{s'} \delta^p_{p'} + \sum_{q \in \mathbb{Z}_5} \frac{\partial G^p}{\partial X^{sp'}_q} D^{(q)}_{ss'} \tag{377}$$

Jacobian:

$$
\begin{aligned}
DF &= \sum_{s=-2}^{N+1} \sum_{s'=-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right| \\
&= \sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right| \\
&\quad + \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right| \\
&\quad + \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right|
\end{aligned}
\tag{378}
$$

$$
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \left[(1-\delta_4^p) \frac{3}{2\,\mathrm{d}t} \delta_{s'}^{s} \delta_{p'}^{p} + \sum_{q\in\mathbb{Z}_5} \frac{\partial G^p}{\partial X_q^{sp'}} D_{ss'}^{(q)} \right] \left|sp\right\rangle \left\langle s'p'\right| \\
&+ \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right| \\
&+ \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle \left\langle s'p'\right|
\end{aligned}
\tag{379}
$$

$$
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N} \sum_{p\in\mathbb{Z}_4} \frac{3}{2\,\mathrm{d}t} \left|sp\right\rangle \left\langle sp\right| \\
&+ \sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \sum_{q\in\mathbb{Z}_5} \frac{\partial G^p}{\partial X_q^{sp'}} D_{ss'}^{(q)} \left|sp\right\rangle \left\langle s'p'\right| \\
&+ \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p}} \left|sp\right\rangle \left\langle s'p\right| \\
&+ \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p}} \left|sp\right\rangle \left\langle s'p\right|
\end{aligned}
\tag{380}
$$

$$
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N}\sum_{p\in\mathbb{Z}_4}\frac{3}{2\,\mathrm{d}t}\left|sp\right\rangle\left\langle sp\right| \\
&+\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2}\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{s,s+h}\left|sp\right\rangle\left\langle s+h,p'\right| \\
&+\sum_{s=-2}^{-1}\sum_{s'=-2}^{1}\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|sp\right\rangle\left\langle s'p\right| \\
&+\sum_{s=N}^{N+1}\sum_{s'=N-2}^{N+1}\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|sp\right\rangle\left\langle s'p\right|
\end{aligned}
\tag{381}
$$

$$
\boxed{
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N}\left|s\right\rangle\left\langle s\right|\otimes\left(\sum_{p\in\mathbb{Z}_4}\frac{3}{2\,\mathrm{d}t}\left|p\right\rangle\left\langle p\right|\right) \\
&+\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2}\left|s\right\rangle\left\langle s+h\right|\otimes\left(\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{s,s+h}\left|p\right\rangle\left\langle p'\right|\right) \\
&+\sum_{s=-2}^{-1}\sum_{s'=-2}^{1}\left|s\right\rangle\left\langle s'\right|\otimes\left(\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|p\right\rangle\left\langle p\right|\right) \\
&+\sum_{s=N}^{N+1}\sum_{s'=N-2}^{N+1}\left|s\right\rangle\left\langle s'\right|\otimes\left(\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|p\right\rangle\left\langle p\right|\right)
\end{aligned}
}
\tag{382}
$$

$$
\boxed{
\begin{gathered}
\left|Y_0\right\rangle = 2\left|X^{t-1}\right\rangle - \left|X^{t-2}\right\rangle \\
\left|F(Y_\alpha)\right\rangle + DF(Y_\alpha)(\left|Y_{\alpha+1}\right\rangle - \left|Y_\alpha\right\rangle)
\end{gathered}
}
\tag{383}
$$

6.1.1 Nonzero p, p' blocks

Let $s \in \mathbb{Z}_N$

$$\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2} |s\rangle\langle s+h| \otimes \left(\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}} D^{(q)}_{s,s+h}|p\rangle\langle p'|\right) \tag{384}$$

$$= \sum_{p,p'\in\mathbb{Z}_5}\sum_{h=-2}^{2}\sum_{q\in\mathbb{Z}_5} D^{(q)}_{0,h}\left[\sum_{s\in\mathbb{Z}_N}|s\rangle\langle s+h|\frac{\partial G^p}{\partial X_q^{sp'}}\right]\otimes|p\rangle\langle p'| \tag{385}$$

so, the diagonal of block (p, p') with offset h is given by

$$\sum_{q\in\mathbb{Z}_5} D^{(q)}_{0,h}\begin{bmatrix} 0 \\ 0 \\ \frac{\partial G^p}{\partial X_q^{0,p'}} \\ \vdots \\ \frac{\partial G^p}{\partial X_q^{sp'}} \\ \vdots \\ \frac{\partial G^p}{\partial X_q^{N,p'}} \\ 0 \\ 0 \end{bmatrix} \tag{386}$$

6.1.2 Frame and centerline (old notation)

$$\boldsymbol{v}^{[0,j]} = \frac{1}{\alpha_1}\partial_s\lambda^{[0,j]}\boldsymbol{e}_1 - \frac{B_2}{\alpha_2}\partial_s^2\omega_3^{[0,j]}\boldsymbol{e}_2 + \frac{B_2}{\alpha_2}\partial_s^2\omega_2^{[0,j]}\boldsymbol{e}_3 \tag{387}$$

$$\boldsymbol{\Omega}^{[0,j]} = \frac{B_1}{\beta}\partial_s\omega_1^{[0,j]}\boldsymbol{e}_1 - \frac{B_2}{\alpha_2}\partial_s^3\omega_2^{[0,j]}\boldsymbol{e}_2 - \frac{B_2}{\alpha_2}\partial_s^3\omega_3^{[0,j]}\boldsymbol{e}_3 \tag{388}$$

$$\psi_t^{[0,j]} = \begin{pmatrix} \hat{\boldsymbol{\Omega}}^{[0,j]} & \boldsymbol{v}^{[0,j]} \\ 0 & 0 \end{pmatrix} \tag{389}$$

$$\psi_s^{[j,k+1]} = \begin{pmatrix} \hat{\boldsymbol{\omega}}^{[j,k+1]} & \boldsymbol{e}_1 \\ 0 & 0 \end{pmatrix} \tag{390}$$

- timestep $t \to t + \mathrm{d}t$

$$\rho^{[0,k+1]} = \rho^{[0,k]} \exp\left(\frac{\mathrm{d}t}{2}\left[\psi_t^{[0,k+1]} + \psi_t^{[0,k]}\right]\right) \tag{391}$$

$$\rho^{[j+1,k+1]} = \rho^{[j,k+1]} \exp\left(\frac{\mathrm{d}s}{2}\left[\psi_s^{[j+1,k+1]} + \psi_s^{[j,k+1]}\right]\right) \tag{392}$$

Required data for timestep $t \to t + \mathrm{d}t$:

- $s = 0$, time t: $Q, x, \partial_s \omega_1, \partial_s^2 \boldsymbol{\omega}_\perp, \partial_s^3 \boldsymbol{\omega}_\perp, \partial_s \lambda$
- $s = 0$, time $t + \mathrm{d}t$: $\partial_s \omega_1, \partial_s^2 \boldsymbol{\omega}_\perp, \partial_s^3 \boldsymbol{\omega}_\perp, \partial_s \lambda$
- all s, time $t + \mathrm{d}t$: $\omega_1, \boldsymbol{\omega}_\perp$
- Parameters: $B_1, B_2, \alpha_1, \alpha_2, \beta$

A Ordered exponentials

Suppose G is a finite dimensional Lie group and let $\mathcal{A}$ be the G-bundle over a compact domain $S \subset \mathbb{R}^p$.

Theorem 1. *Let g, h be sections of $\mathcal{A}$, which satisfy $g(s^1, \ldots, s^p) = g(0, \ldots, 0, s^k, 0 \ldots, 0) h(s^1, \ldots, s^p)$ and let $\psi_g = \psi_{g,n} \mathrm{d}s^n$, $\psi_h = \psi_{h,n} \mathrm{d}s^n$ be the Maurer-Cartan forms for g, h. Then*

$$h(0, \ldots, 0, s^k, 0 \ldots, 0) = 1 \tag{393}$$

$$\psi_{g,j} = \begin{cases} \psi_{h,j} & \textit{for } j \neq k \\ \psi_{h,k} + h^{-1} \psi_{g,k}(\ldots, 0, s^k, 0 \ldots) h & \textit{for } j = k \end{cases} \tag{394}$$

$$\psi_{h,k}(0, \ldots, 0, s^k, 0 \ldots, 0) = 0 \tag{395}$$

$$g(s) = g(0)\, Oexp\left[\psi_{g,k}(0, \ldots, s^k, \ldots, 0); s^k\right] h(s) \tag{396}$$

Proof. Equation (393) follows from evaluating the defining eqn of h at $s = (0, \ldots, 0, s^k, 0 \ldots, 0)$. Equation (394) follows from taking derivatives, and Equation (395) follows from evaluating Equation (394) at $s = (0, \ldots, 0, s^k, 0 \ldots, 0)$ and using Equation (393). □

Theorem 2. *Let g be a section of $\mathcal{A}$ with Maurer-Cartan form $\psi = \psi_n \, \mathrm{d}s^n$. Then,*

$$g(s) = g(0) \, Oexp\left[\psi_1(\cdot, 0 \ldots); s^1\right] \, Oexp\left[\psi_2(s^1, \cdot, 0 \ldots); s^2\right] \cdots Oexp\left[\psi_p(s^1, \ldots, s^{p-1}, \cdot); s^p\right] \tag{397}$$

$$= g(0) \prod_{k=1}^{p} Oexp\left[\psi_1(s^1, \cdots, s^{k-1}, \cdot, 0 \ldots); s^k\right] \tag{398}$$

Proof. By induction... □

Example 1. *Suppose g represents the state of a Kirchoff rod, so, $S \subset \mathbb{R}^{1+1}$, $G = SE(3)$. Given the Maurer Cartan form and the initial configuration of a single cross-section, we can reconstruct the configuration of a Kirchoff rod at later times by solving a sequence of initial value problems. Define $h(s,t)$ to be the transformation which maps cross-section 0 onto cross-section s at time t.*

$$g(s,t) = g(0,t)h(s,t) \tag{399}$$

$$h(0,t) = 1 \tag{400}$$

$$\mathrm{d}g =: g\psi_g \tag{401}$$

$$\mathrm{d}h =: h\psi_h \tag{402}$$

$$\psi_{g,s}(s,t) = \psi_{h,s}(s,t) \tag{403}$$

$$\psi_{g,t}(s,t) = \psi_{h,t}(s,t) + h^{-1}(s,t)\psi_{g,t}(0,t)h(s,t) \tag{404}$$

$$\psi_{h,t}(0,t) = 0 \tag{405}$$

$$h(s,t) = Oexp\left[\psi_{g,s}(\cdot, t); s\right] \tag{406}$$

$$g(0,t) = g(0,0) \, Oexp\left[\psi_{g,t}(0, \cdot); t\right] \tag{407}$$

$$g(s,t) = g(0,0) \, Oexp\left[\psi_{g,t}(0, \cdot); t\right] \, Oexp\left[\psi_{g,s}(\cdot, t); s\right] \tag{408}$$

$$\psi_s = \begin{pmatrix} \hat{\omega} & e_1 \\ 0 & 0 \end{pmatrix} \tag{409}$$

$$\psi_t = \begin{pmatrix} \hat{\Omega} & v \\ 0 & 0 \end{pmatrix} \tag{410}$$

A.1 References

- [3]

B Finite difference formulas

$$\boxed{\frac{3\omega^{[k+1]} - 4\omega^{[k]} + \omega^{[k-1]}}{2\,\mathrm{d}t} = \frac{\partial \omega}{\partial t}^{[k+1]} - \frac{1}{3}\,\mathrm{d}t^2\,\frac{\partial^3 \omega}{\partial t^3}^{[k+1]} + O(\mathrm{d}t^3)} \tag{411}$$

Proof.

$$\begin{aligned} -4\omega(t+h-h) &= -4\left(\omega(t+h) - h\dot{\omega}(t+h) + \frac{1}{2}h^2\ddot{\omega}(t+h) - \frac{1}{6}h^3\dddot{\omega}(t+h)\right) + O(h^4) \\ &= -4\omega(t+h) + 4h\dot{\omega}(t+h) - 2h^2\ddot{\omega}(t+h) + \frac{2}{3}h^3\dddot{\omega}(t+h) + O(h^4) \end{aligned}$$

$$\omega(t+h-2h) = \omega(t+h) - 2h\dot{\omega}(t+h) + 2h^2\ddot{\omega}(t+h) - \frac{4}{3}h^3\dddot{\omega}(t+h) + O(h^4)$$

$$-4\omega(t+h-h) + \omega(t+h-2h) = -3\omega(t+h) + 2h\dot{\omega}(t+h) + 2h^2\ddot{\omega}(t+h) - \frac{2}{3}h^3\dddot{\omega}(t+h) + O(h^4)$$

$$\frac{3\omega(t+h) - 4\omega(t+h-h) + \omega(t+h-2h)}{2h} = \dot{\omega}(t+h) - \frac{1}{3}h^2\dddot{\omega}(t+h) + O(h^3)$$

□

$$\boxed{\frac{\omega^{[j+1,k+1]} - 2\omega^{[j,k+1]} + \omega^{[j-1,k+1]}}{\mathrm{d}s^2} = \frac{\partial^2 \omega}{\partial s^2}^{[j,k+1]} + \frac{1}{12}\,\mathrm{d}s^2\,\omega^{[j,k+1]''''} + O(\mathrm{d}s^3)} \tag{412}$$

Proof.

$$\omega(s+h) = \omega(s) + h\omega'(s) + \frac{1}{2}h^2\omega''(s) + \frac{1}{6}h^3\omega'''(s) + \frac{1}{24}h^4\omega''''(s) + O(h^5)$$

$$\omega(s-h) = \omega(s) - h\omega'(s) + \frac{1}{2}h^2\omega''(s) - \frac{1}{6}h^3\omega'''(s) + \frac{1}{24}h^4\omega''''(s) + O(h^5)$$

$$\omega(s+h) + \omega(s-h) = 2\omega(s) + h^2\omega''(s) + \frac{1}{12}h^4\omega''''(s) + O(h^5)$$

$$\frac{\omega(s+h) - 2\omega(s) + \omega(s-h)}{h^2} = \omega''(s) + \frac{1}{12}h^2\omega''''(s) + O(h^3)$$

□

$$\boxed{\omega^{[k+1]} = 2\omega^{[k]} - \omega^{[k-1]} + \mathrm{d}t^2\,\ddot{\omega}(t) + O(\mathrm{d}t^4)} \tag{413}$$

Proof.

$$\frac{\omega(t+h) - 2\omega(t) + \omega(t-h)}{h^2} = \ddot{\omega}(t) + O(h^2)$$

$$2\omega(t) - \omega(t-h) = \omega(t+h) - h^2\ddot{\omega}(t) + O(h^4)$$

□

$$\boxed{\rho(s)\exp\left(\frac{h}{2}\left[\psi(s) + \psi(s+h)\right]\right) = \rho(s+h) + \frac{h^3}{12}\rho(s)\left(\psi'' - \psi\psi' + \psi'\psi\right) + O(h^4)} \tag{414}$$

Proof.

$$\begin{aligned}
\rho' &= \rho\psi \\
\rho'' &= \rho\left(\psi' + \psi^2\right) \\
\rho''' &= \rho\left(\psi'' + \psi'\psi + 2\psi\psi' + \psi^3\right)
\end{aligned}$$

$$\begin{aligned}
\rho(s+h) &= \rho(s) + h\rho'(s) + \frac{1}{2}h^2\rho''(s) + \frac{1}{6}h^3\rho'''(s) + O(h^4) \\
&= \rho\left[1 + h\psi + \frac{1}{2}h^2\left(\psi' + \psi^2\right) + \frac{1}{6}h^3\left(\psi'' + \psi'\psi + 2\psi\psi' + \psi^3\right)\right] + O(h^4)
\end{aligned}$$

$$\begin{aligned}
\exp\left(\frac{h}{2}\left[\psi(s) + \psi(s+h)\right]\right) &= \exp\left(\frac{h}{2}\left[2\psi(s) + h\psi'(s) + \frac{1}{2}h^2\psi''(s) + O(h^3)\right]\right) \\
&= \exp\left(h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + O(h^4)\right) \\
&= 1 + h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + \frac{1}{2}\left[h\psi + \frac{1}{2}h^2\psi'\right]^2 + \frac{1}{6}\left[h\psi\right]^3 + O(h^4) \\
&= 1 + h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + \frac{1}{2}\left[h^2\psi^2 + \frac{1}{2}h^3\psi\psi' + \frac{1}{2}h^3\psi'\psi\right] + \frac{1}{6}h^3\psi^3 + O(h \\
&= 1 + h\psi + \frac{1}{2}h^2\left[\psi' + \psi^2\right] + h^3\left[\frac{1}{4}\psi'' + \frac{1}{4}\psi\psi' + \frac{1}{4}\psi'\psi + \frac{1}{6}\psi^3\right] + O(h^4)
\end{aligned}$$

$$\begin{aligned}
\rho(s)\exp\left(\frac{h}{2}\left[\psi(s) + \psi(s+h)\right]\right) &= \rho(s)\left(1 + h\psi + \frac{1}{2}h^2\left[\psi' + \psi^2\right] + h^3\left[\frac{1}{4}\psi'' + \frac{1}{4}\psi\psi' + \frac{1}{4}\psi'\psi + \frac{1}{6}\psi^3\right]\right) + \\
&= \rho(s+h) + h^3\rho(s)\left(\frac{1}{4}\psi'' + \frac{1}{4}\psi\psi' + \frac{1}{4}\psi'\psi + \frac{1}{6}\psi^3\right) \\
&\quad - h^3\rho(s)\left(\frac{1}{6}\psi'' + \frac{1}{6}\psi'\psi + \frac{1}{3}\psi\psi' + \frac{1}{6}\psi^3\right) + O(h^4) \\
&= \rho(s+h) + \frac{h^3}{12}\rho(s)\left(\psi'' - \psi\psi' + \psi'\psi\right) + O(h^4)
\end{aligned}$$

□

C Biharmonic stuff

C.1 Green function

$$\partial_s^4 G(s,s') = \delta(s-s') \tag{415}$$
$$G(0,s') = G(1,s') = 0 \tag{416}$$
$$\partial_s G(s,s')|_{s=0} = \partial_s G(s,s')|_{s=1} = 0 \tag{417}$$

$$G(x,y) = \begin{cases} \frac{1}{12}(1-x)(1-y)^2x^2y + \frac{1}{24}(y-x)(1-y)^2x^2, & \text{for } x<y \\ \frac{1}{12}(1-y)(1-x)^2y^2x + \frac{1}{24}(x-y)(1-x)^2y^2, & \text{for } y<x \end{cases} \tag{418}$$

$$p(x,y) = \frac{1}{12}(1-x)(1-y)^2x^2y + \frac{1}{24}(y-x)(1-y)^2x^2 \tag{419}$$

$$G(x,y) = p(x,y)\Theta(y-x) + p(y,x)\Theta(x-y) \tag{420}$$

C.2 Free-free eigenfunctions

$$\partial_s^4 \psi_k = k^4 \psi_k \tag{421}$$

$$\psi_k = a_1\cos(ks) + a_2\sin(ks) + a_3\cosh(ks) + a_4\sinh(ks) \tag{422}$$

$$\partial_s\psi_k = -ka_1\sin(ks) + ka_2\cos(ks) + ka_3\sinh(ks) + ka_4\cosh(ks) \tag{423}$$

$$\psi_k(0) = \psi_k(1) = 0, \qquad \partial_s\psi_k(0) = \partial_s\psi_k(1) = 0 \tag{424}$$

$$\psi_k = [\sinh(k) - \sin(k)]\,[\cos(ks) - \cosh(ks)] + [\cos(k) - \cosh(k)]\,[\sin(ks) - \sinh(ks)] \tag{425}$$

$$\cos(k)\cosh(k) = 1 \tag{426}$$

$$k_0 \approx 4.73004074 \tag{427}$$
$$k_1 \approx 7.85320462 \tag{428}$$
$$k_2 \approx 10.99560784 \tag{429}$$
$$k_3 \approx 14.13716549 \tag{430}$$
$$k_4 \approx 17.27875966 \tag{431}$$
$$k_5 \approx 20.42035225 \tag{432}$$
$$k_6 \approx 23.5619449 \tag{433}$$
$$k_7 \approx 26.70353756 \tag{434}$$
$$k_8 \approx 29.84513021 \tag{435}$$
$$k_9 \approx 32.98672286 \tag{436}$$
$$k_{10} \approx 36.12831552 \tag{437}$$
$$k_{11} \approx 39.26990817 \tag{438}$$
$$k_{12} \approx 42.41150082 \tag{439}$$
$$k_{13} \approx 45.55309348 \tag{440}$$
$$k_{14} \approx 48.69468613 \tag{441}$$
$$k_{15} \approx 51.83627878 \tag{442}$$
$$k_{16} \approx 54.97787144 \tag{443}$$
$$k_{17} \approx 58.11946409 \tag{444}$$
$$k_{18} \approx 61.26105675 \tag{445}$$
$$k_{19} \approx 64.4026494 \tag{446}$$

$$k_0/\pi \approx 1.50561873 \tag{447}$$
$$k_1/\pi \approx 2.49975267 \tag{448}$$
$$k_2/\pi \approx 3.50001068 \tag{449}$$
$$k_3/\pi \approx 4.49999954 \tag{450}$$
$$k_4/\pi \approx 5.50000002 \tag{451}$$
$$k_5/\pi \approx 6.5 \tag{452}$$
$$k_6/\pi \approx 7.5 \tag{453}$$
$$k_7/\pi \approx 8.5 \tag{454}$$
$$k_8/\pi \approx 9.5 \tag{455}$$
$$k_9/\pi \approx 10.5 \tag{456}$$
$$k_{10}/\pi \approx 11.5 \tag{457}$$
$$k_{11}/\pi \approx 12.5 \tag{458}$$
$$k_{12}/\pi \approx 13.5 \tag{459}$$
$$k_{13}/\pi \approx 14.5 \tag{460}$$
$$k_{14}/\pi \approx 15.5 \tag{461}$$
$$k_{15}/\pi \approx 16.5 \tag{462}$$
$$k_{16}/\pi \approx 17.5 \tag{463}$$
$$k_{17}/\pi \approx 18.5 \tag{464}$$
$$k_{18}/\pi \approx 19.5 \tag{465}$$
$$k_{19}/\pi \approx 20.5 \tag{466}$$

Proof.

$$\psi_k(0) = a_1 + a_3 = 0, \qquad \partial_s \psi_k(0) = ka_2 + ka_4 = 0 \tag{467}$$

$$\psi_k = a_1 \left[\cos(ks) - \cosh(ks)\right] + a_2 \left[\sin(ks) - \sinh(ks)\right] \tag{468}$$

$$\partial_s \psi_k = -ka_1 \left[\sin(ks) + \sinh(ks)\right] + ka_2 \left[\cos(ks) - \cosh(ks)\right] \tag{469}$$

$$\psi_k(1) = a_1 \left[\cos(k) - \cosh(k)\right] + a_2 \left[\sin(k) - \sinh(k)\right] = 0 \tag{470}$$

$$\psi_k = -\left[\sin(k) - \sinh(k)\right]\left[\cos(ks) - \cosh(ks)\right] + \left[\cos(k) - \cosh(k)\right]\left[\sin(ks) - \sinh(ks)\right] \tag{471}$$

$$\partial\psi_k = k\left[\sin(k) - \sinh(k)\right]\left[\sin(ks) + \sinh(ks)\right] + k\left[\cos(k) - \cosh(k)\right]\left[\cos(ks) - \cosh(ks)\right] \tag{472}$$

$$\partial\psi_k(1) = k\left[\sin(k) - \sinh(k)\right]\left[\sin(k) + \sinh(k)\right] + k\left[\cos(k) - \cosh(k)\right]\left[\cos(k) - \cosh(k)\right] \tag{473}$$

$$= 2k\left[1 - \cos(k)\cosh(k)\right] \tag{474}$$

$$\cos(k)\cosh(k) = 1 \tag{475}$$

□

C.2.1 Free-free alt rep

$$\Psi_+ = \cosh(k/2)\cos(k[s-1/2]) - \cos(k/2)\cosh(k[s-1/2]) \tag{476}$$

$$\sin(k/2) + \cos(k/2)\tanh(k/2) = 0 \tag{477}$$

$$k_0, k_2, k_4, \ldots \tag{478}$$

$$\langle\Psi_+|\Psi_+\rangle = \frac{1}{2}\left[\cosh^2(k/2) + \cos^2(k/2)\right] \tag{479}$$

Normalized

$$\psi_+ = \frac{\sqrt{2}}{\sqrt{1 + \frac{\cos^2(k/2)}{\cosh^2(k/2)}}}\cos(k(s-1/2)) - \frac{\sqrt{2}\cos(k/2)}{(1+e^{-k})\sqrt{1 + \frac{\cos^2(k/2)}{\cosh^2(k/2)}}}\left(e^{k(s-1)} + e^{-ks}\right) \tag{480}$$

$$\Psi_- = -\sinh(k/2)\sin(k[s-1/2]) + \sin(k/2)\sinh(k[s-1/2]) \tag{481}$$

$$-\sin(k/2) + \cos(k/2)\tanh(k/2) = 0 \tag{482}$$

$$k_1, k_3, k_5, \ldots \tag{483}$$

$$\langle \Psi_- | \Psi_- \rangle = \frac{1}{2} \left[\sinh^2(k/2) - \sin^2(k/2) \right] \tag{484}$$

Normalized

$$\psi_- = \frac{-\sqrt{2}}{\sqrt{1 - \frac{\sin^2(k/2)}{\sinh^2(k/2)}}} \sin(k(s - 1/2)) + \frac{\sqrt{2}\sin(k/2)}{(1 - e^{-k})\sqrt{1 - \frac{\sin^2(k/2)}{\sinh^2(k/2)}}} \left(e^{k(s-1)} - e^{-ks} \right) \tag{485}$$

C.2.2 Asymptotics

$$\frac{\cos(ks) - \cosh(ks)}{\cos(k) - \cosh(k)} + \frac{\sin(ks) - \sinh(ks)}{\sinh(k) - \sin(k)} \tag{486}$$

$$\cos(k) \sim 0 \tag{487}$$

$$k_n = (j + 3/2)\pi + \varepsilon_j \tag{488}$$

D Green function and some spectral stuff

Let $\hat{H}$ be a local self adjoint operator.

$$\langle x | \hat{H} | x' \rangle = H(x)\delta(x - x') \tag{489}$$

$$\hat{H}^\dagger = \hat{H} \tag{490}$$

Let $|k\rangle$ represent the eigenvector of $\hat{H}$ with eigenvalue k.

$$\hat{H} |k\rangle = k |k\rangle \tag{491}$$

The right inverse of $\hat{H}$ is called a Green function

$$\hat{H}\hat{G} = 1 \tag{492}$$

$$\langle x | \hat{G} | x' \rangle = G(x, x') \tag{493}$$

$$\langle x | \hat{H}\hat{G} | x' \rangle = H(x)G(x, x')\delta(x - x') \tag{494}$$

$$\hat{G} = \sum_k \frac{1}{k} |k\rangle \langle k| \tag{495}$$

E Laplace/Poisson equation stuff

E.0.1 Dirichlet eigenfunctions

Let $0 < s < 1$ and $n = 1, 2, 3, \ldots$

$$\langle s|\phi_n\rangle = \phi_n(s) = \sqrt{2}\sin(n\pi s) \tag{496}$$

$$\langle \phi_m|\phi_n\rangle = \int_0^1 \phi_m(s)\phi_n(s)\,\mathrm{d}s = \delta_{mn} \tag{497}$$

$$\partial_s^2\phi_n = -(n\pi)^2\phi_n \tag{498}$$

The eigenfunctions can also be separated into even/odd functions about $s = 1/2$ representation. Let $m = 1, 2, 3, \ldots$

$$|\phi_{2m}\rangle := (-1)^m \left|\phi_m^-\right\rangle \tag{499}$$

$$|\phi_{2m-1}\rangle := (-1)^{m-1} \left|\phi_m^+\right\rangle \tag{500}$$

$$\phi_m^+(s) = \sqrt{2}\cos((2m-1)\pi(s-1/2)) \tag{501}$$

$$\phi_m^-(s) = \sqrt{2}\cos(2m\pi(s-1/2)) \tag{502}$$

Note the parity of the index is opposite the parity of the functions!

Proof.

$$\sin(n\pi s) = \sin(n\pi(s-1/2) + n\pi/2) \tag{503}$$

$$= \cos(n\pi/2)\sin(n\pi(s-1/2)) + \sin(n\pi/2)\cos(n\pi(s-1/2)) \tag{504}$$

$$= \begin{cases} \cos(n\pi/2)\sin(n\pi(s-1/2)), & n \text{ even} \\ \sin(n\pi/2)\cos(n\pi(s-1/2)), & n \text{ odd} \end{cases} \tag{505}$$

For $n = 2m, m = n/2$ we have

$$\cos(n\pi/2) = \cos(m\pi) = (-1)^m = (-1)^{n/2} \tag{506}$$

$$\sin(n\pi s) = (-1)^{n/2}\sin(n\pi(s-1/2)). \tag{507}$$

and for $n = 2m - 1, m = (n+1)/2$ we have

$$\sin(n\pi/2) = \sin((2m-1)\pi/2) = (-1)^{m-1} = (-1)^{(n-1)/2} \tag{508}$$

$$\sin(n\pi s) = (-1)^{(n-1)/2} \cos(n\pi(s - 1/2)). \tag{509}$$

$$\sin(n\pi s) = \begin{cases} (-1)^{n/2} \sin(n\pi(s-1/2)), & n \text{ even} \\ (-1)^{(n-1)/2} \cos(n\pi(s-1/2)), & n \text{ odd} \end{cases} \tag{510}$$

□

E.0.2 Dirichlet Green function

Let $s, s' \in (0,1)$.

$$\partial_s^2 G_{s'} = \delta_{s'} \tag{511}$$

$$G_{s'}(0) = G_{s'}(1) = 0 \tag{512}$$

$$G_{s'}(s) = \begin{cases} (s'-1)s, & \text{for } s < s' \\ s'(s-1), & \text{for } s > s' \end{cases} \tag{513}$$

$$= ss' - s\Theta(s'-s) - s'\Theta(s-s') \tag{514}$$

Proof.

$$G_z(s) = \begin{cases} a_- s + b_-, & \text{for } s < z \\ a_+ s + b_+, & \text{for } s > z \end{cases} \tag{515}$$

$$G_z(0) = 0 \Rightarrow b_- = 0 \tag{516}$$

$$G_z(1) = 0 \Rightarrow b_+ = -a_+ \tag{517}$$

$$G_z(z^+) = G_z(z^-) \Rightarrow a_+ = z, a_- = (z-1) \tag{518}$$

□

Fourier space representation

$$\hat{G} = \sum_{n=1}^{\infty} -\frac{1}{n^2\pi^2} \left|\phi_n\right\rangle \left\langle\phi_n\right| \tag{519}$$

$$G(s,s') = \left\langle s\right| \hat{G} \left|s'\right\rangle = \sum_{n=1}^{\infty} -\frac{2}{n^2\pi^2} \sin(n\pi s)\sin(n\pi s') \tag{520}$$

F Some Fourier stuff

$$\sin(x+y) = \sin(x)\cos(y) + \sin(y)\cos(x) \tag{521}$$

$$\cos(x+y) = \cos(x)\cos(y) - \sin(y)\sin(x) \tag{522}$$

$$\langle\psi|\phi\rangle = \int_0^1 \psi^*(x)\phi(x)\,\mathrm{d}x \tag{523}$$

$$\theta_p(x) = \cos(p(x-1/2)) \tag{524}$$

$$\phi_p(x) = \sin(p(x-1/2)) \tag{525}$$

$$\mu_p(x) = \cosh(p(x-1/2)) \tag{526}$$

$$\nu_p(x) = \sinh(p(x-1/2)) \tag{527}$$

$$\langle\theta_p|\theta_q\rangle = 2\frac{p\sin(p/2)\cos(q/2) - q\cos(p/2)\sin(q/2)}{p^2-q^2} \tag{528}$$

$$\langle\theta_p|\mu_q\rangle = 2\frac{p\sin(p/2)\cosh(q/2) + q\cos(p/2)\sinh(q/2)}{p^2+q^2} \tag{529}$$

$$\langle\phi_p|\phi_q\rangle = 2\frac{q\sin(p/2)\cos(q/2) - p\cos(p/2)\sin(q/2)}{p^2-q^2} \tag{530}$$

$$\langle \phi_p | \nu_q \rangle = 2 \frac{q \sin(p/2) \cosh(q/2) - p \cos(p/2) \sinh(q/2)}{p^2 + q^2} \tag{531}$$

$$\langle \theta_p | \phi_q \rangle = \langle \theta_p | \nu_q \rangle = \langle \phi_p | \mu_q \rangle = \langle \mu_p | \nu_q \rangle = 0 \tag{532}$$

$$\langle \phi_p | \phi_q \rangle = 2 \frac{q \sin(p/2) \cos(q/2) - p \cos(p/2) \sin(q/2)}{p^2 - q^2} \tag{533}$$

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